Kinematics

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Motion in a Straight Line

Kinematics is the study of motion of bodies without considering the factors causing the motion.

Kinematics is broadly studied in three parts:

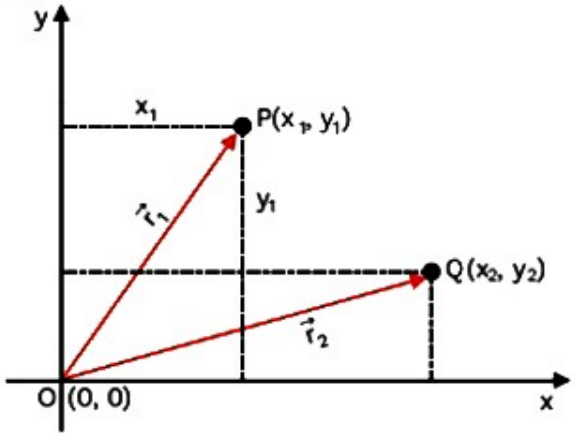
1. Motion in a straight line ⇒ Motion in one dimension
2. Motion in a plane ⇒ Motion in 2-dimensions
3. Relative motion

MOTION IN A STRAIGHT LINE :

Let us define few terms related to straight line motion.

Position :

- It is a vector quantity.
- We use cartesian plane to represent position. Co-ordinates of point P and Q represent their positions.



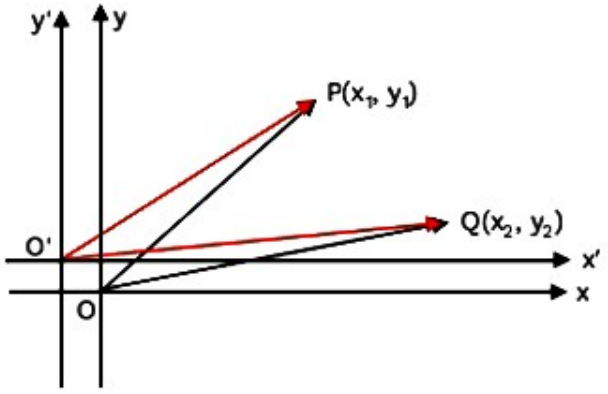
$\overrightarrow{OP}$ = Position vector of point P

$\overrightarrow{OP} = x_1\hat{i} + y_1\hat{j} = \vec{r}_1$

$\overrightarrow{OQ}$ = Position vector of point Q

$\overrightarrow{OQ} = x_2\hat{i} + y_2\hat{j} = \vec{r}_2$

If cartesian plane is changed.



$\overrightarrow{O'P} \neq x_1\hat{i} + y_1\hat{j}$

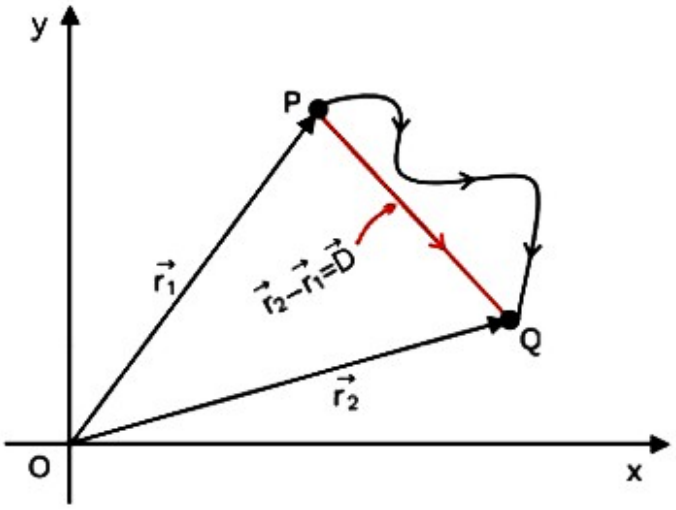
$\overrightarrow{O'P} = x'_1\,\hat{i} + y'_1\,\hat{j}$

∴ On changing cartesian plane, position vector changes.
Position vector depends on choice of cartesian plane.

Displacement :

- Change of position vector is known as displacement.

- It is a vector quantity.
- Displacement = Change in position vector.



Suppose an insect moves from point 'P' to point 'Q'.

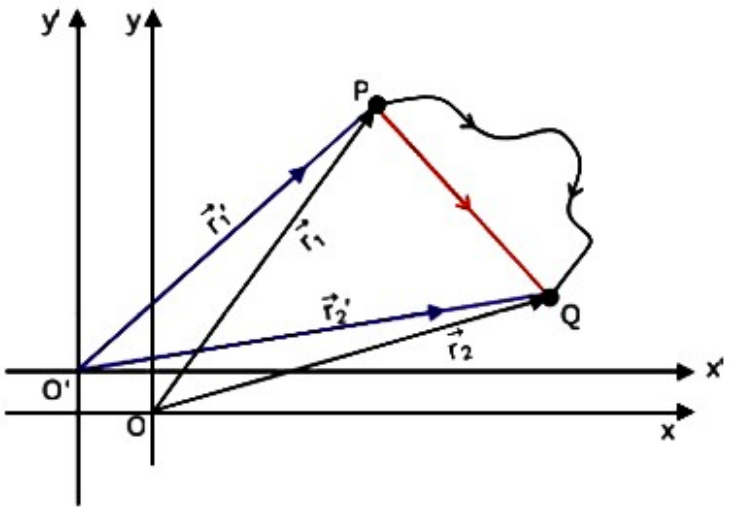
Initial position of insect, $\vec{r}_i = \vec{r}_1 = \overrightarrow{OP}$

Final position of insect, $\vec{r}_f = \vec{r}_2 = \overrightarrow{OQ}$

Displacement, $\vec{D} = \vec{r}_f - \vec{r}_i$

$$\boxed{\vec{D} = \vec{r}_2 - \vec{r}_1}$$

$$\vec{D} = \overrightarrow{OQ} - \overrightarrow{OP} = \overrightarrow{PQ}$$

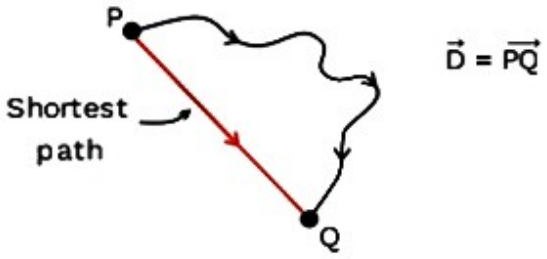


- Direction of displacement is from initial to final position.
- If cartesian plane is changed

  $\vec{D} = \vec{r_2} - \vec{r_1} = \vec{r'_2} - \vec{r'_1}$

  $\overrightarrow{OQ} - \overrightarrow{OP} = \overrightarrow{O'Q} - \overrightarrow{O'P}$

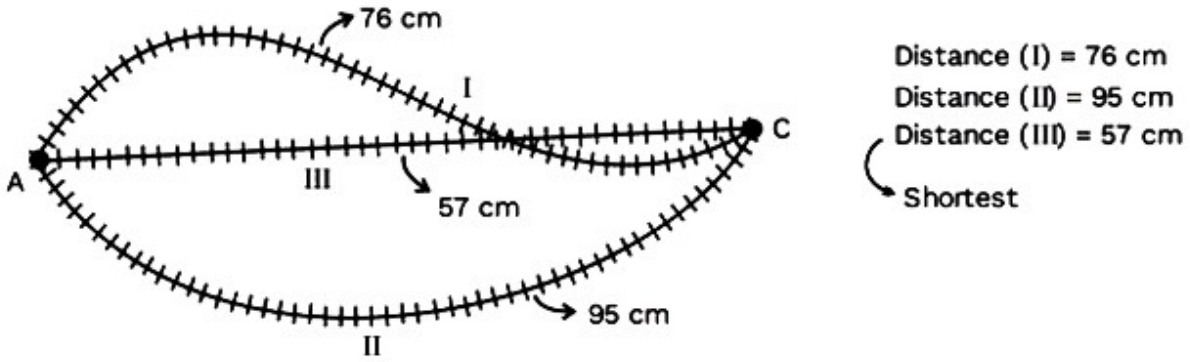
  ∴ Displacement remains same after changing cartesian plane.



Magnitude of displacement = Length of shortest path = $|\vec{D}|$

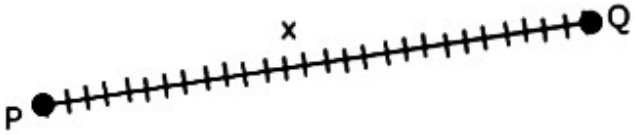
Distance :

- It is also called as path length.



- Distance between two points can have many values depending on path taken.
- It is a scalar quantity.
- Distance ≥ d_0 (shortest path)

  ∴ Distance ≥ |Displacement|
- For a straight line path



Distance = |Displacement|

$|\vec{D}| = x$

- For a very small path

Distance = |Displacement|

Velocity :

- It is rate of change of displacement with respect to time,
  or Rate of change of position vector w.r.t time

$$\vec{V} = \frac{\overrightarrow{\Delta D}}{\Delta t} = \frac{\overrightarrow{\Delta x}}{\Delta t}$$

Small interval ($\Delta t \to 0$)

$$\vec{V} = \frac{\overrightarrow{\Delta x}}{\Delta t} = \frac{\vec{dx}}{dt}$$

Instantaneous velocity

Big interval

$$V = \frac{\overrightarrow{\Delta x}}{\Delta t}$$

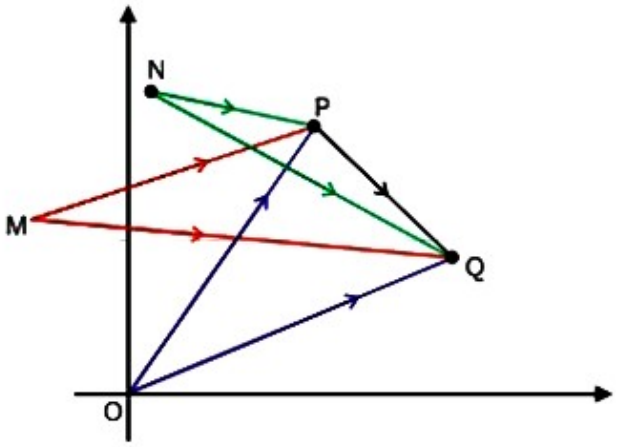
Average velocity

- Velocity is a vector quantity.

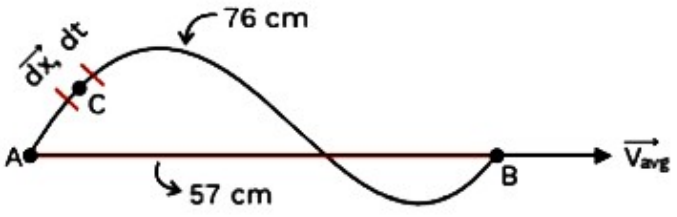
$$\vec{V} = \frac{\overrightarrow{MQ} - \overrightarrow{MP}}{\Delta t} = \frac{\overrightarrow{\Delta D}}{\Delta t} = \frac{\overrightarrow{PQ}}{\Delta t}$$

$$\vec{V} = \frac{\overrightarrow{OQ} - \overrightarrow{OP}}{\Delta t} = \frac{\overrightarrow{PQ}}{\Delta t}$$

$$\vec{V} = \frac{\overrightarrow{NQ} - \overrightarrow{NP}}{\Delta t} = \frac{\overrightarrow{PQ}}{\Delta t}$$



- If time taken to reach from A to B is 4 sec.



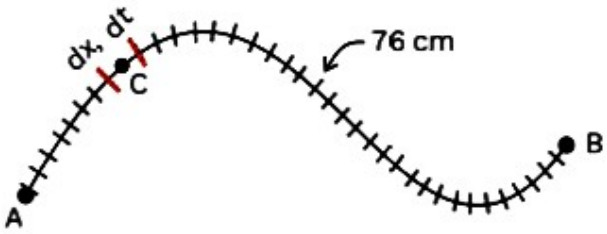
Average velocity from A to B $= V = \frac{\vec{\Delta x}}{\Delta t} = \frac{57}{4}$ cm / sec along $\overrightarrow{AB}$

Instantaneous velocity at C, $\overrightarrow{V_c} = \frac{d\vec{x}}{dt} \Rightarrow |\overrightarrow{V_c}| = \frac{dx}{dt}$

Speed :

- Speed $= \frac{\text{Path length}}{\text{time}} = \frac{\text{Distance}}{\text{time}}$

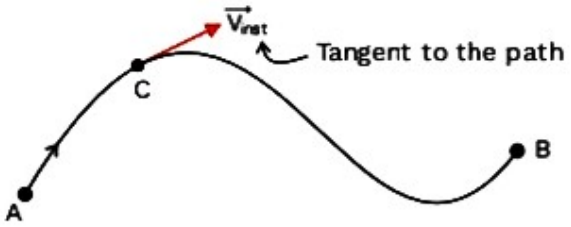
If time taken to reach B from A = 4s



- Average speed $= \frac{\text{total path length}}{\text{total time}} = \frac{76}{4} = 19 \text{ cm/sec}$
- Instantaneous speed at C $= \frac{\text{small distance}}{\text{small time}} = \frac{dx}{dt} = V$
- Instantaneous speed = |Instantaneous velocity|

$$\frac{dx}{dt} = \left|\frac{d\vec{x}}{dt}\right|$$

- Average speed ≥ |Average velocity|
- **Direction of instantaneous velocity**



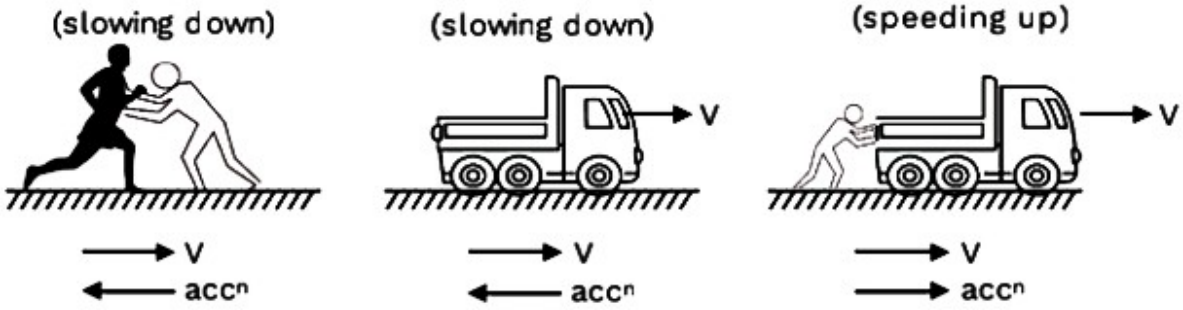
Direction of instantaneous velocity is along the tangent drawn to the point on the path.

Acceleration :

- It is rate of change of velocity w.r.t. time.

$$\vec{a} = \frac{d\vec{V}}{dt} \qquad \vec{a} = \frac{\vec{V_2} - \vec{V_1}}{t_2 - t_1}$$

- When velocity and acceleration have same direction, then body speeds up.
- When velocity and acceleration have opposite direction, then body slows down.



Equations of motion :

$V = u + at$

$S = ut + \frac{1}{2}at^2$

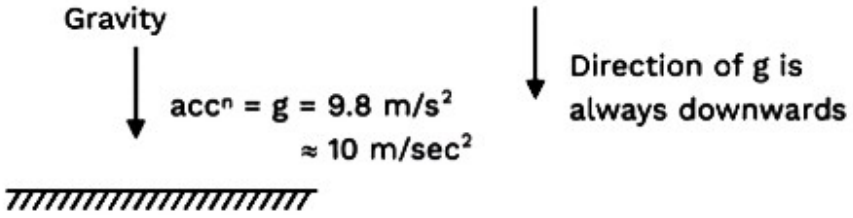
$V^2 = u^2 + 2aS$

V = Final velocity
u = Initial velocity
a = Acceleration
S = Displacement
All are taken with sign

Valid when
(1) Motion is along straight line
(2) Acceleration is constant

Application of equation of motion

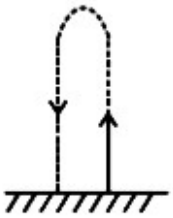
- Free fall → motion under gravity only.

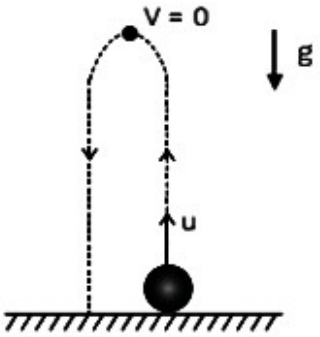


(1) Time to go up, $t = \frac{u}{g}$

Time to go down, $t = \frac{u}{g}$

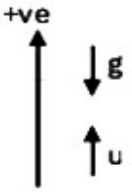
Total time, $\boxed{t = \frac{2u}{g}}$





Proof :

For upward journey,



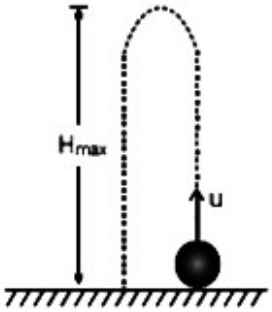
At topmost point, V = 0

$V = u + at$

$\Rightarrow \quad 0 = +u + (-g)t$

$\Rightarrow \quad t = \frac{u}{g}$

(2) Maximum height attained, $\boxed{H_{max} = \frac{u^2}{2g}}$



Proof :

At topmost point, V = 0

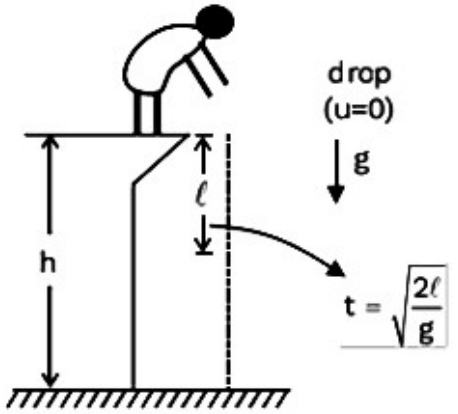


$u = +u, \quad a = -g, \quad S = +H_{max}$

$V^2 - u^2 = 2aS$

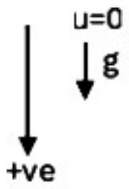
$\Rightarrow \quad 0^2 - u^2 = 2(-g)H_{max} \quad \Rightarrow \quad H_{max} = \frac{u^2}{2g}$

(3)



$\boxed{t = \sqrt{\frac{2h}{g}}}$ → Time to reach the ground

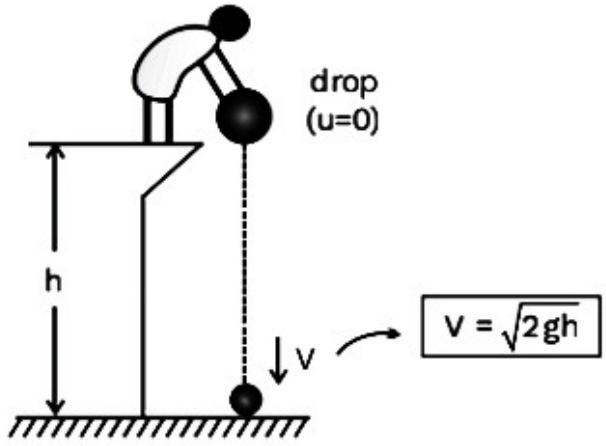
Proof :



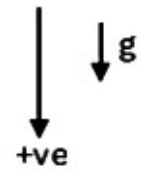
u = 0, a = +g, S = +h

$$S = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad h = (0 \times t) + \frac{1}{2}g\,t^2 \quad \Rightarrow \quad t = \sqrt{\frac{2h}{g}}$$

(4)



Proof :



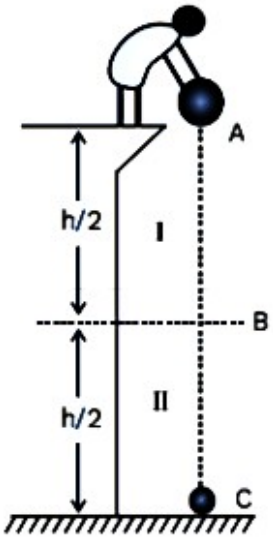
u = 0, a = +g, S= +h, V = ?

$V^2 - u^2 = 2aS$

$\Rightarrow \quad V^2 - 0^2 = 2gh$

$\Rightarrow \quad V = \sqrt{2gh}$

Q. **Find $\frac{t_I}{t_{II}}$ or $\frac{t_{AB}}{t_{BC}}$ if ball is being dropped from rest at A.**



Sol. Ball will move faster from B to C as compared to A to B.

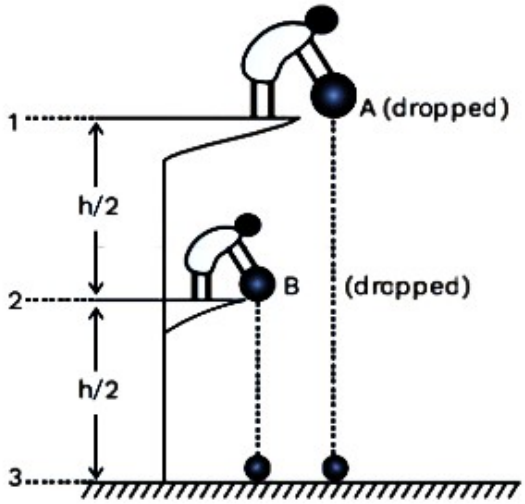
$\therefore \quad t_{BC} < t_{AB}$

$$t_{AB} = \sqrt{\frac{2 \cdot \frac{h}{2}}{g}} = \sqrt{\frac{h}{g}} = t_I \quad , \quad t_{AC} = \sqrt{\frac{2h}{g}}$$

$$t_{BC} = t_{AC} - t_{AB} = \sqrt{\frac{2h}{g}} - \sqrt{\frac{h}{g}} = t_{II}$$

$$\frac{t_I}{t_{II}} = \frac{\sqrt{\frac{h}{g}}}{\sqrt{\frac{2h}{g}} - \sqrt{\frac{h}{g}}} = \frac{\sqrt{\frac{h}{g}}}{\sqrt{\frac{h}{g}}(\sqrt{2} - 1)} = \frac{1}{\sqrt{2} - 1}$$

Q. **For h = 20 m, first ball 'A' is dropped and when it reaches near ball 'B', ball B is dropped at that instant. Find the difference in time of sound produced when ball 'A' and ball 'B' hit the ground.**



Sol.

$$t_{12} = \sqrt{\frac{2\times 10}{10}} = \sqrt{2}\ \text{sec}$$

$$t_{13} = \sqrt{\frac{2\times 20}{10}} = 2\ \text{sec}$$

$$t_{23} = t_{13} - t_{12} = (2-\sqrt{2})\ \text{sec}$$

} Ball 'A'

$$t_{23} = \sqrt{\frac{2\times 10}{10}} = \sqrt{2}\ \text{sec}$$

} Ball 'B'

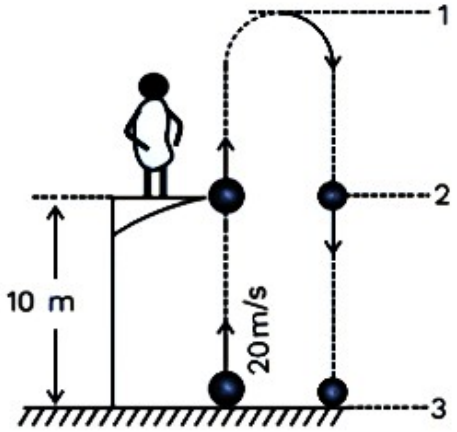
Ball 'A' takes $(2-\sqrt{2})$ sec to reach from point 2 to 3 and,

Ball 'B' takes $\sqrt{2}$ sec to reach from point 2 to 3

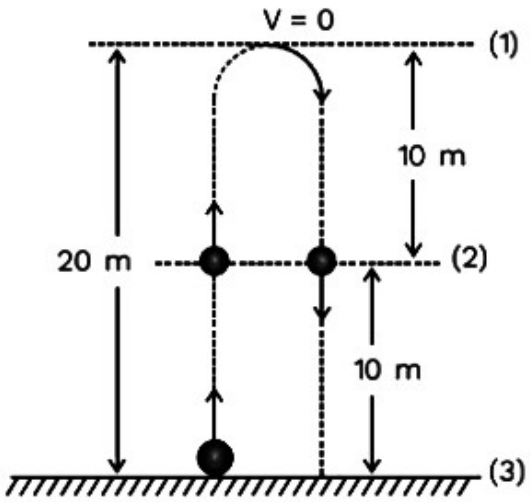
$\therefore$ Time difference in sound produced.

$\Delta t = \sqrt{2} - (2-\sqrt{2}) = (2\sqrt{2}-2)$ sec

Q. **A person sees the ball at position (2) twice, once when ball is moving up and once when ball is going down. Find the difference in time between these two events.**



Sol.



Maximum height attained by ball,

$$H_{max} = \frac{20 \times 20}{2 \times 10} = 20\,m$$

Time taken by ball to reach point 2 from point 1.

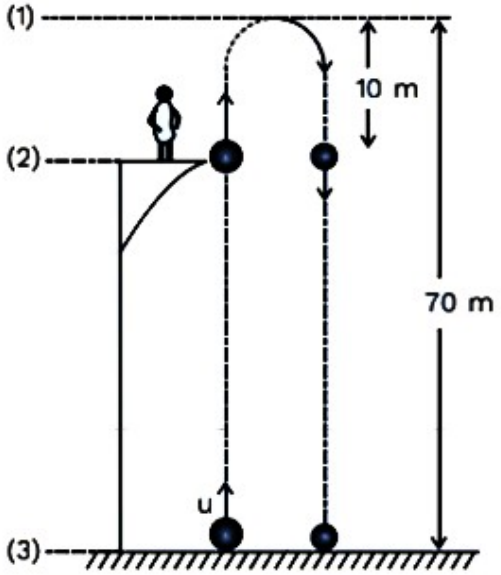
$$t_{12} = \sqrt{\frac{2 \times 10}{10}} = \sqrt{2}\ \text{sec}$$

$$t_{12} = t_{21}$$

$\therefore$ Time difference, $\Delta t = \sqrt{2} + \sqrt{2}$

$= 2\sqrt{2}$ sec

Q. **If ball reaches a maximum height of 70 m, find u.**
Also find the time gap between the moments when person standing at position (2) sees the ball.



Sol. $H_{max} = \frac{u^2}{2g} \Rightarrow 70 = \frac{u^2}{2 \times 10}$

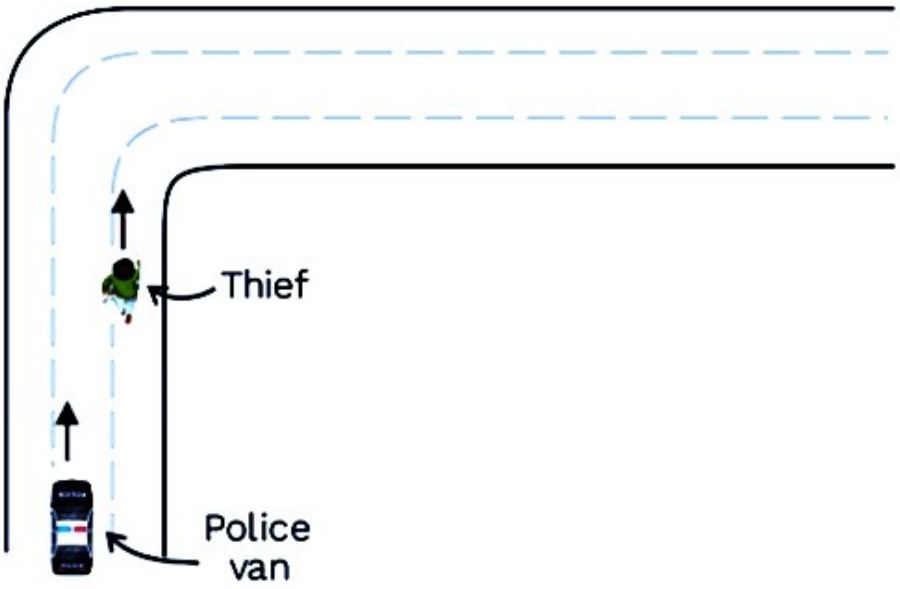
$\Rightarrow u = 10\sqrt{14}$ m / s

For time gap,

$$\Delta t = t_{12} + t_{21} = 2t_{12} = 2\sqrt{\frac{2 \times 10}{10}}$$

$\Delta t = 2\sqrt{2}$ sec

Q.



Speed of thief, V_T = 72 km/hr

Speed of police van, V_P = 90 km/hr

First thief takes turn and after 10 sec police van also takes turn.

Find how much time after thief taking turn, police will catch the thief.

Sol.

$$V_T = 72 \times \frac{5}{18} = 20 \text{ m/s}$$

$$V_P = 90 \times \frac{5}{18} = 25 \text{ m/s}$$

Let t seconds after thief taking turn, police catches him.

$\therefore$ Distance travelled by thief after taking turn, D = 20 × t

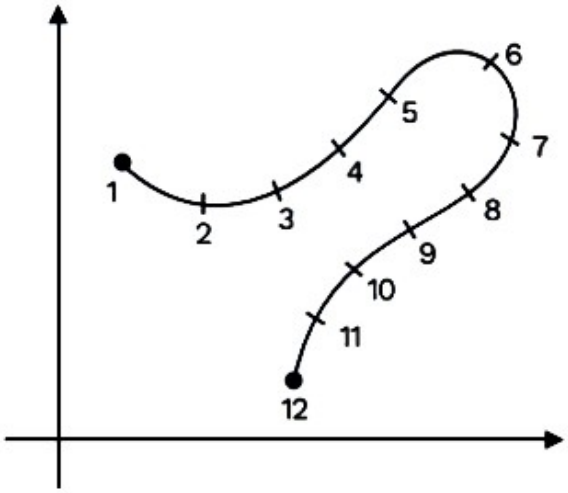
Distance travelled by police after taking turn, D = 25 × (t − 10)

$20t = 25(t - 10)$

$\Rightarrow$ $4t = 5t - 50$

$\Rightarrow$ $t = 50$ sec

Q. **A particle moves on a curved path as shown. It starts from point as shown. It starts from point 1 and ends at point 12.**
Taking initial position as point 1, find the location where direction of average velocity and instantaneous velocity are same.

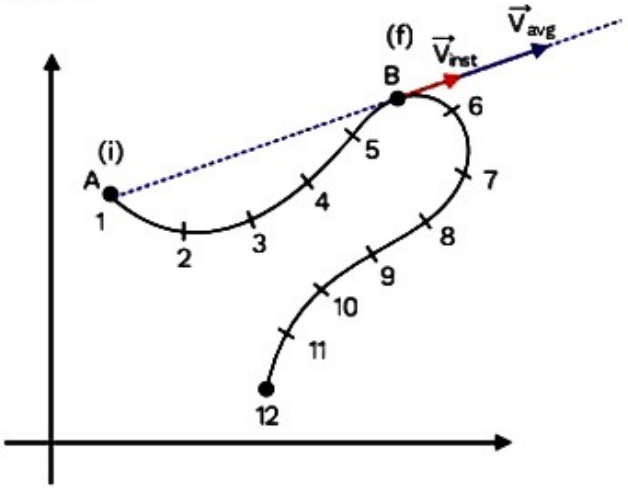


Sol. Location lies between points 5 and 6

Direction of $\overrightarrow{V_{avg}}$ = Direction of displacement

Direction of $\overrightarrow{V_{inst}}$ = Direction of tangent

Direction of tangent drawn at point B is same as direction of displacement between point A and B.



Variable acceleration :

- Acceleration is not constant.
  - 3 equations of motion are not applicable.
  - Calculus is used.
- x = f(t) is given

$$x \xrightarrow{\text{differentiate}} \frac{dx}{dt} = V \xrightarrow{\text{differentiate}} \frac{dV}{dt} = \frac{d^2x}{dt^2} = a$$

- a = f(t) is given

$$\frac{dV}{dt} = a \Rightarrow \int dV = \int a\,dt \Rightarrow v = \int a\,dt$$

- $a = \frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt} = V\frac{dV}{dx}$

$$\Rightarrow \boxed{a = V\frac{dV}{dx}}$$

Q. **Acceleration of a particle is given by, $a = 3t^2 + 2t + 2$. If initial velocity of particle, u = 2 m/s and at t = 0, particle is at x = 0. Find velocity of particle at t = 2 sec.**

Sol.

$$\frac{dV}{dt} = 3t^2 + 2t + 2$$

$$\Rightarrow \int_0^t dV = \int_0^t 3t^2 dt + \int_0^t 2t\,dt + \int_0^t 2\,dt$$

$$\Rightarrow [V]_2^V = \left[t^3\right]_0^t + \left[t^2\right]_0^t + [2t]_0^t$$

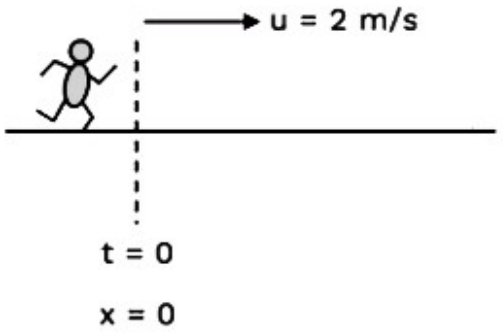
$$\Rightarrow V - 2 = t^3 + t^2 + 2t$$

$$\Rightarrow V = t^3 + t^2 + 2t + 2$$

at t = 2 sec

$$\Rightarrow V = 2^3 + 2^2 + (2 \times 2) + 2$$

$$\Rightarrow V = 18 \text{ m/s}$$



Q. **The variation of position of a particle moving in the xy-plane with time t is given by the equation,**
$x = (3t^2 - 6t)$ m, $y = (t^2 - 2t)$ m.
State the correct statements about the particle?

(A) **Acceleration of the particle is zero at t = 0 sec.**
(B) **Velocity of the particle is zero at t = 0 sec.**
(C) **Velocity of the particle is zero at t = 1 sec.**
(D) **Velocity and acceleration of the particle never become zero.**

Sol. Position at t = 0

$t = 0 \Rightarrow x = 3 \times 0^2 - 2 \times 0 = 0$

$t = 0 \Rightarrow y = 0^2 - 2 \times 0 = 0$

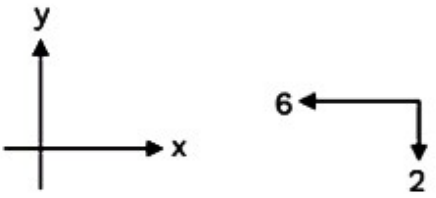
$\therefore$ Particle is at origin at t = 0

Velocity in x direction, $V_x = \frac{dx}{dt} = 3 \times 2t - 6 = 6t - 6$

Velocity in y-direction, $V_y = \frac{dy}{dx} = 2 \times t - 2 = 2t - 2$

At t = 0,

$V_x = -6$ m/s, $V_y = -2$ m/s



$\therefore \quad V = \sqrt{(-6)^2 + (-2)^2} = \sqrt{40}$

So, V is non-zero at t = 0

Acceleration in x-direction, $a_x = \frac{dV_x}{dt} = \frac{d^2x}{dt^2} = 6$

Acceleration in y-direction, $a_y = \frac{dV_y}{dt} = \frac{d^2y}{dt^2} = 2$

$\vec{a} = 6\hat{i} + 2\hat{j}$

$|\vec{a}| = \sqrt{6^2 + 2^2} = \sqrt{40}$

Acceleration is constant (not changing with time). Acceleration is never zero.

At t = 1 sec,

$V_x = 6 \times 1 - 6 = 0$ and $V_y = 2 \times 1 - 2 = 0$

$V = \sqrt{V_x^2 + V_y^2} = \sqrt{0^2 + 0^2}$

V = 0

$\therefore$ Velocity is zero at t = 1 sec

Option 'C' is correct.

Q. **Acceleration of a particle moving along x-axis is given by a = −4x (where x is in m and a is in m/s²). At t = 0, particle passes through origin with 2 m/s velocity. Find**

(A) Velocity as a function of x ⇒ V = f(x)

(B) Position as a function of time ⇒ x = f(t).

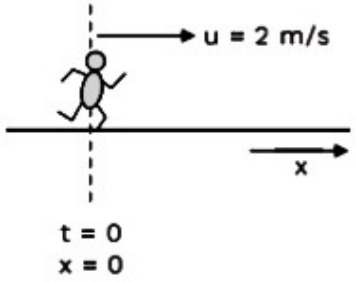
Sol. $a = \frac{V\,dV}{dx} \Rightarrow \frac{V\,dV}{dx} = -4x \Rightarrow \int_2^V V dV = -\int_0^x 4x dx$ [∵ at x = 0, u = 2 m/s]

$\Rightarrow \left[\frac{V^2}{2}\right]_2^V = -4\left[\frac{x^2}{2}\right]_0^x \Rightarrow \frac{V^2}{2} - \frac{2^2}{2} = -4\left[\frac{x^2}{2} - \frac{0^2}{2}\right]$

$\Rightarrow V^2 - 4 = -4x^2$

$\Rightarrow V^2 = 4 - 4x^2$

$\Rightarrow V = \sqrt{4 - 4x^2}$



Variable acceleration

$\Rightarrow \boxed{V = 2\sqrt{1 - x^2}}$ ...(1) Ans.

We need, x = f(t)

$\Rightarrow \frac{dx}{dt} = 2\sqrt{1 - x^2}$

$\Rightarrow \int_0^x \frac{dx}{\sqrt{1 - x^2}} = \int_0^t 2\,dt \quad \left[\int \frac{dx}{\sqrt{1 - x^2}} = \sin^{-1} x + c\right]$

$\Rightarrow \left[\sin^{-1} x\right]_0^x = 2\left[t\right]_0^t$

$\Rightarrow \sin^{-1} x - \sin^{-1} 0 = 2(t - 0)$

$\Rightarrow \sin^{-1} x = 2t$

$\Rightarrow \boxed{x = \sin 2t}$ ...(2) Ans.

Q. **A particle moves along x-axis with acceleration given by, $a = (3 - 2t)$ m/s². At $t = 0$ sec and at $t = 6$ sec, particle is at same position. Then find**

(A) Initial velocity of particle.

(B) Velocity of particle at $t = 2$ sec.

Sol.

$$a = \frac{dV}{dt} = 3 - 2t$$

$$\Rightarrow \int_{V_0}^{V} dV = \int_{0}^{t} (3 - 2t)\,dt$$

$$\Rightarrow [V]_{V_0}^{V} = \left[3t - t^2\right]_0^t = \left[(3t - t^2) - (0)\right] = 3t - t^2$$

$$\Rightarrow \quad V = V_0 + 3t - t^2,$$

where, V_0 is initial velocity i.e. velocity at $t = 0$

Now,

$$V = \frac{dx}{dt}$$

$$\Rightarrow \int dx = \int V dt \Rightarrow \int_{x_0}^{x} dx = \int_{0}^{t} (V_0 + 3t - t^2)dt \Rightarrow [x]_{x_0}^{x} = \left[V_0 t + \frac{3t^2}{2} - \frac{t^3}{3}\right]_0^t$$

$$\Rightarrow x - x_0 = V_0 t + \frac{3t^2}{2} - \frac{t^3}{3} \Rightarrow x = x_0 + V_0 t + \frac{3t^2}{2} - \frac{t^3}{3},$$

Where x_0 is initial position

i.e., at $t = 0$, $x = x_0$

Now given that particle's position is same at $t = 0$ and $t = 6$ sec.

$\therefore$ At $t = 0$ and $t = 6$ sec, $x = x_0$

$$\Rightarrow x_0 = x_0 + (V_0 \times 6) + 3 \times \frac{6^2}{2} - \frac{6^3}{3}$$

$$\Rightarrow 6V_0 = 72 - 54 = 18$$

$$\Rightarrow V_0 = 3 \text{ m/s}$$

$\therefore$ Initial velocity, $V_0 = 3$ m/s ...(1) Ans.

Now,

$V = V_0 + 3t - t^2$

$V = 3 + 3t - t^2$

At $t = 2$ sec,

$V = 3 + 3 \times 2 - 2^2$

$V = 5$ m/s

$\therefore$ Velocity at $t = 2$ sec, $V = 5$ m/s ...(2) Ans.

Q. **Relation between time 't' and position 'x' is given as $t = ax^2 + bx$, where a, b are constants. Find retardation.**

Sol. $t = \alpha x^2 + \beta x$

$$\Rightarrow \frac{dt}{dx} = \alpha \times 2x + \beta \times 1 \Rightarrow \frac{dx}{dt} = \frac{1}{\left(\frac{dt}{dx}\right)} = \frac{1}{2\alpha x + \beta} = V$$

$$\text{Now, acceleration} = \frac{dV}{dt} = \frac{d^2x}{dt} = \frac{(2\alpha x + \beta)\cdot \frac{d1}{dt} - 1 \cdot \frac{d}{dt}(2\alpha x + \beta)}{(2\alpha x + \beta)^2}$$

$$\text{Acceleration} = \frac{(2\alpha x + \beta) \times 0 - \left(2\alpha \times \frac{dx}{dt}\right)}{(2\alpha x + \beta)^2}$$

$$\text{Acceleration} = \frac{1}{(2\alpha x + \beta)^2}(-2\alpha)\left(\frac{dx}{dt}\right) = V^2(-2\alpha)V$$

Acceleration $= -2\alpha V^3$

$\therefore$ Retardation $= 2\alpha V^3$

Graphical representation of motion:

Two types of questions are asked.

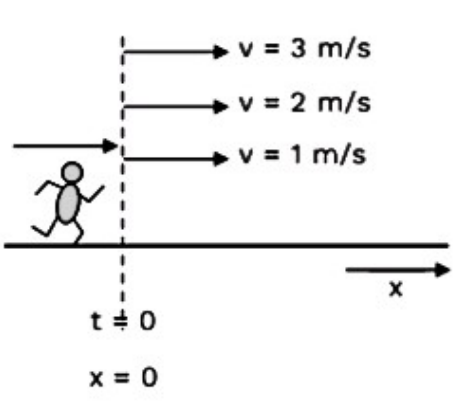
How to read the graph.

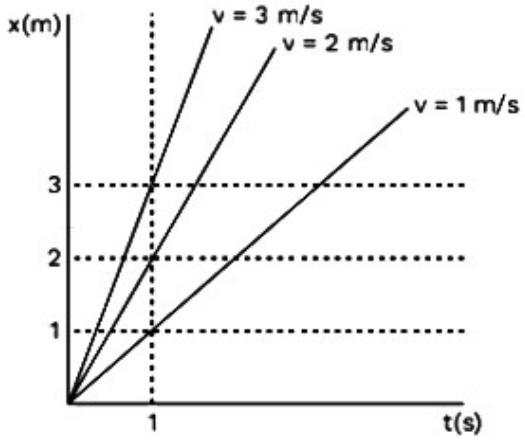
How to draw the graph.

In this section, we will see about graphical representation of motion.

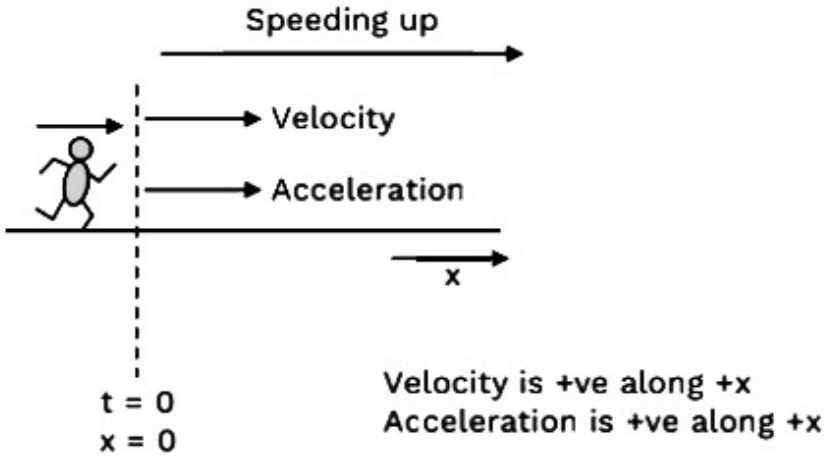
Position-time graph

Lets say a person is running on a straight horizontal road. Different speeds of the person are shown in figure and position time graph has been plotted.

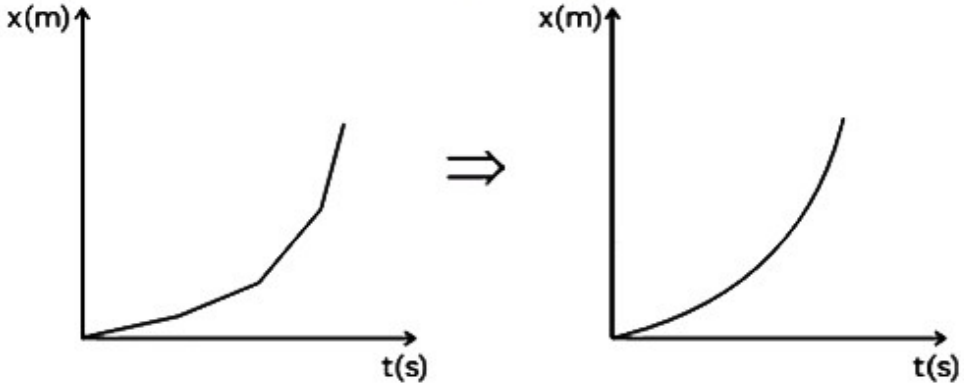




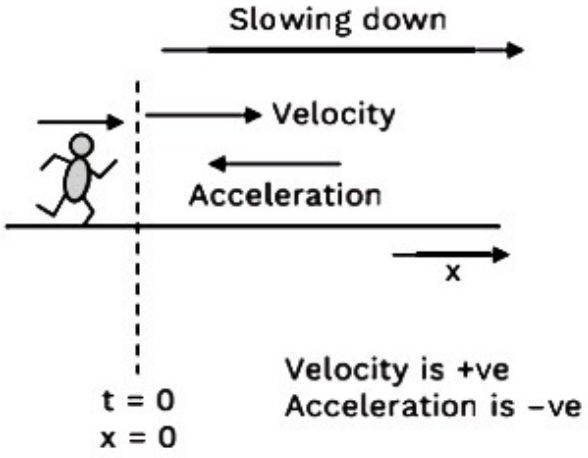
As velocity of person increases, slope of position versus time graph increases.
Now, lets consider the person is speeding up as shown.



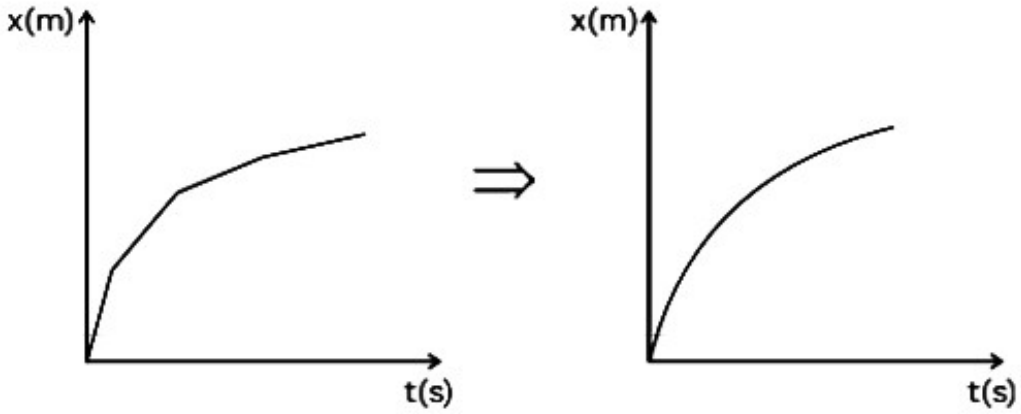
We can draw this variation for very small time intervals as shown. On making the time interval infinitely small, we get a smooth upward parabolic curve for x-t variation.



Now, if the person is slowing down as shown below-



The velocity will decrease with time. As a result, slope of x-t graph must decrease in very small time intervals. On making the time intervals infinitely small, we get a smooth rightward parabola.



Velocity-time graph :

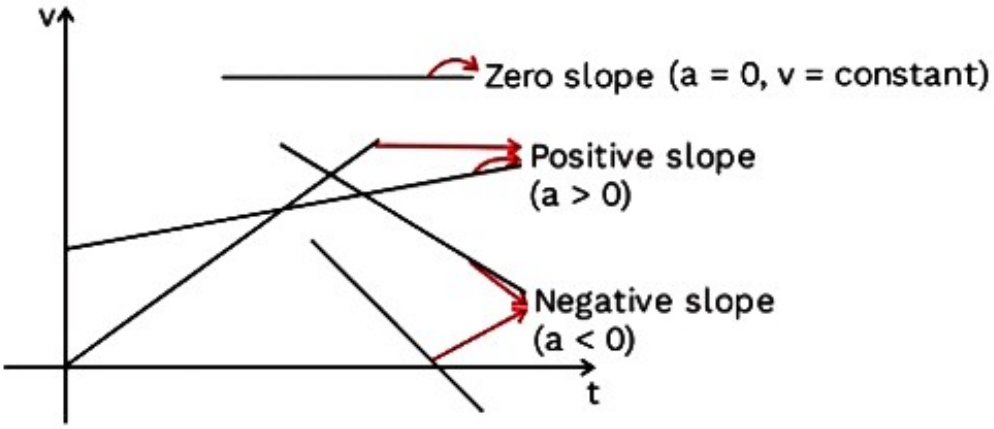
If acceleration is constant.

$a = \frac{dV}{dt}$ = constant

$a = \frac{dV}{dt}$ = slope of V-t graph

a = constant ⇒ slope of V-t graph is constant

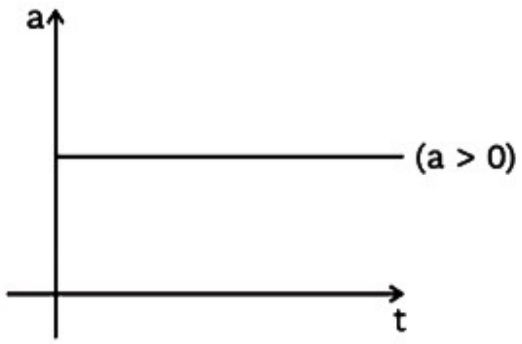
Slope constant ⇒ Straight line

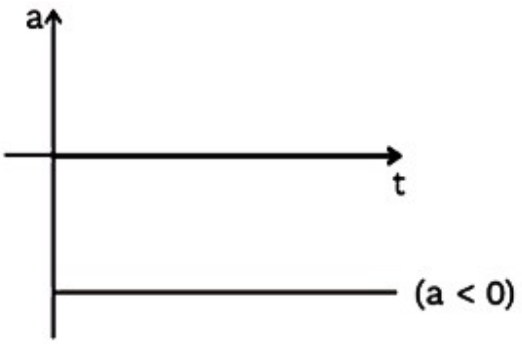


a-t graph :

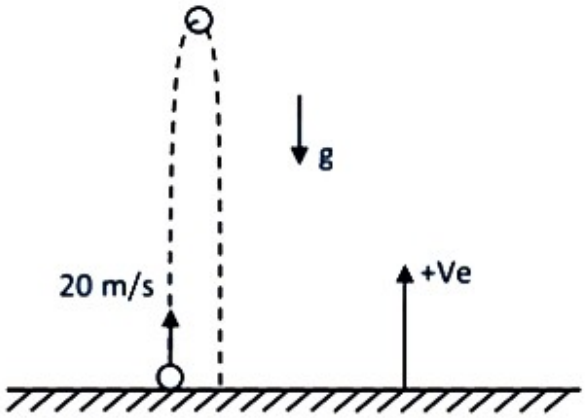
If acceleration is constant, then a-t graph is a straight line parallel to t-axis.

a = constant

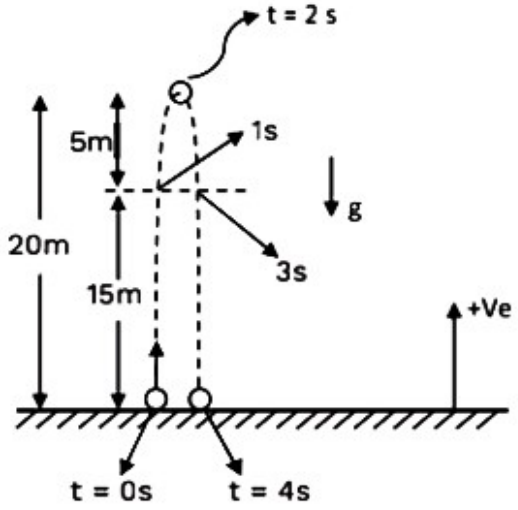


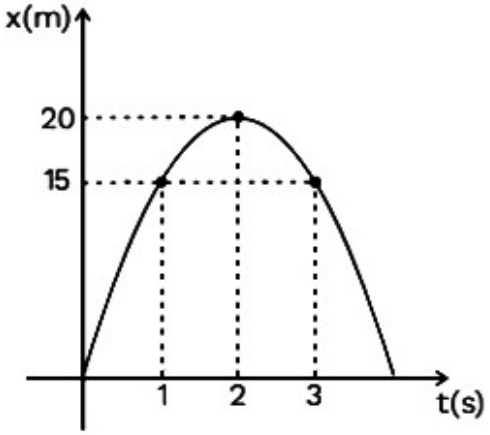


Q. **Taking upward direction as positive, Plot x vs t graph, where x is displacement of ball from ground.**

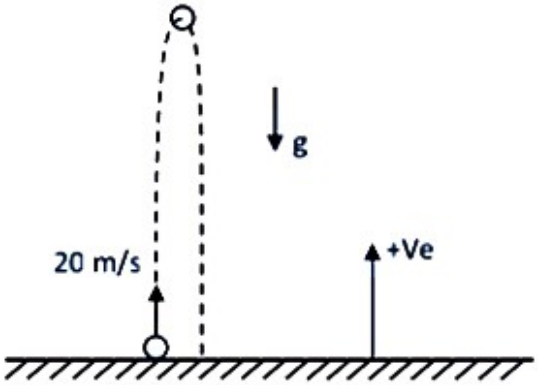


Sol. Maximum height, $H_{max} = \frac{u^2}{2g} = \frac{20 \times 20}{2 \times 10} = 20$ m





Q. **Draw velocity-time and speed-time graph**



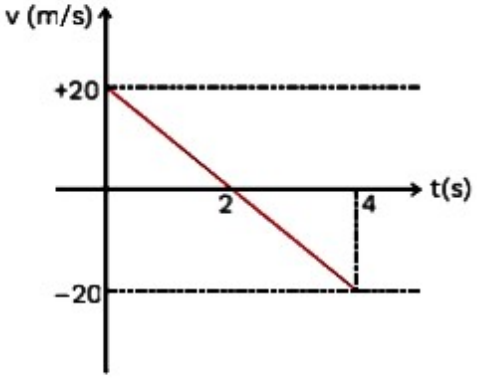
Sol. $a = \frac{dv}{dt} = -g \quad \Rightarrow \quad \frac{dv}{dt} = -10$

Here slope of v-t graph is constant (constant slope means straight line) and negative

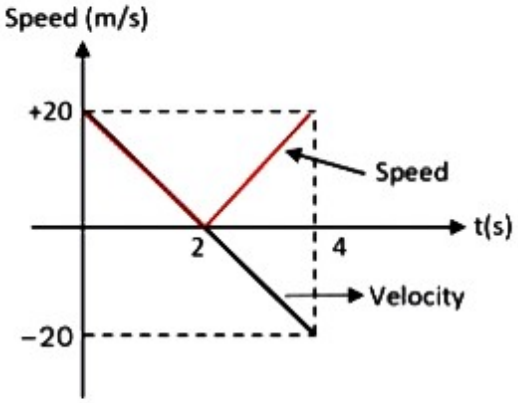
At t = 0, u = +20

At t = 2s, u = 0

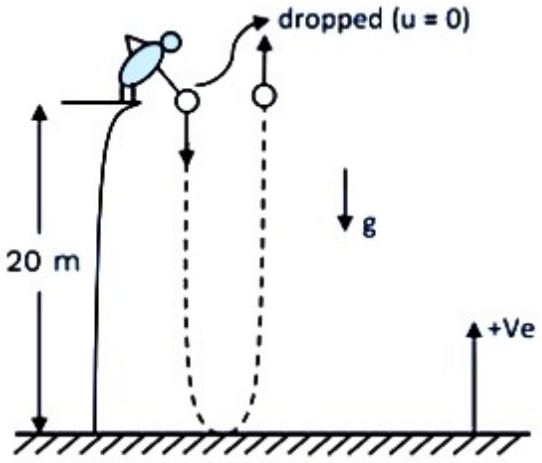
At t = 4 sec, u = –20



Speed vs time graph
Speed = |Velocity|
|+ve| = +ve
|−ve| = +ve



Q. **Taking upward direction as positive, draw velocity-time graph and displacement-time graph where displacement is measured from ground.**



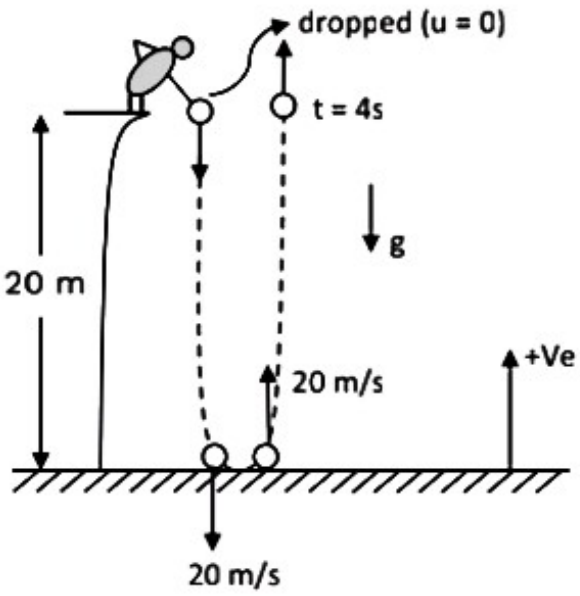
Sol. $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 20}{10}} = 2\,s$

$V = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = 20$ m/s

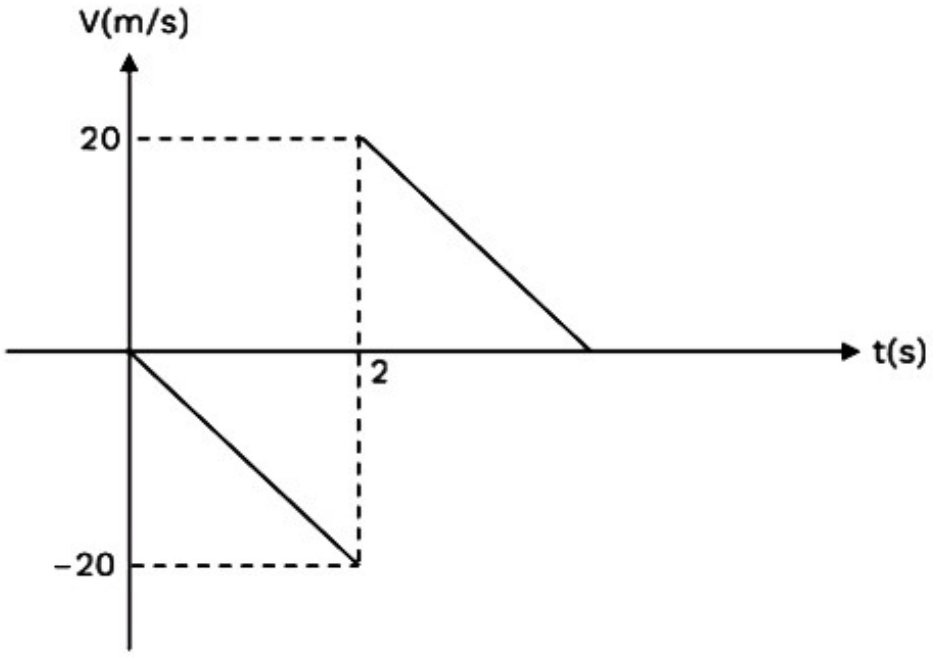
$a = \frac{dV}{dt} = -g = -10 =$ constant

$\because \quad \frac{dV}{dt} =$ constant

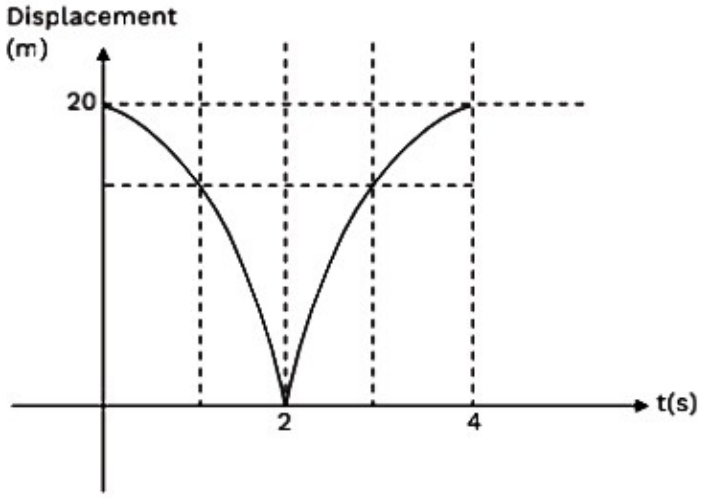
$\therefore$ V-t graph is straight line.



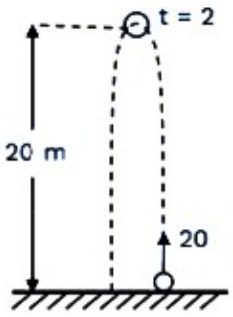
Velocity-time graph



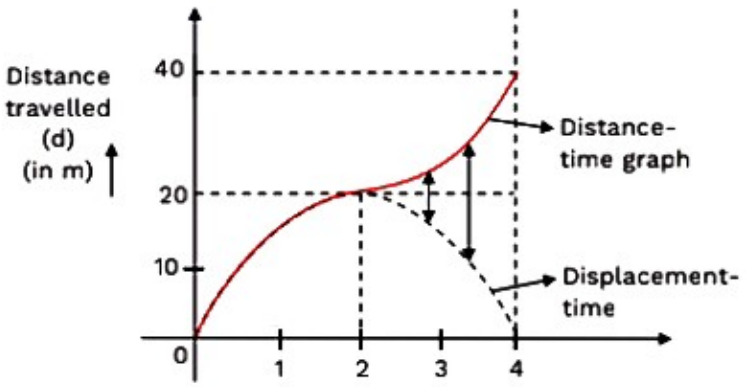
Displacement-time graph



Q. **Draw the distance time graph for given case.**



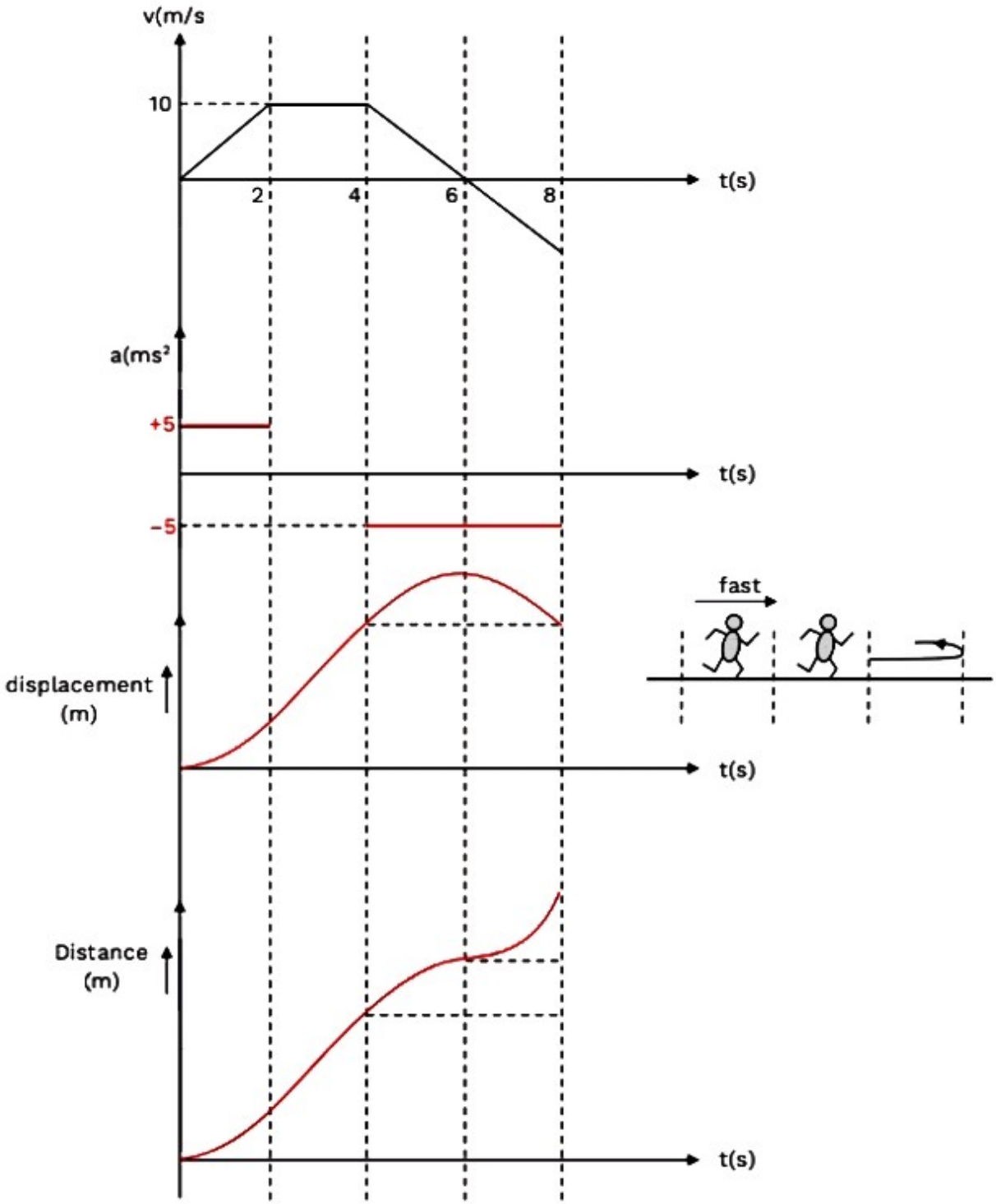
Sol.



Distance travelled is always positive and it can never decrease.

Q. **For given velocity-time graph, draw a-t, displacement-t, distance-t graphs.**

Sol.



$$a = \frac{dv}{dt} = \text{slope of v–t curve}$$

from t = 0 to t = 2 sec,

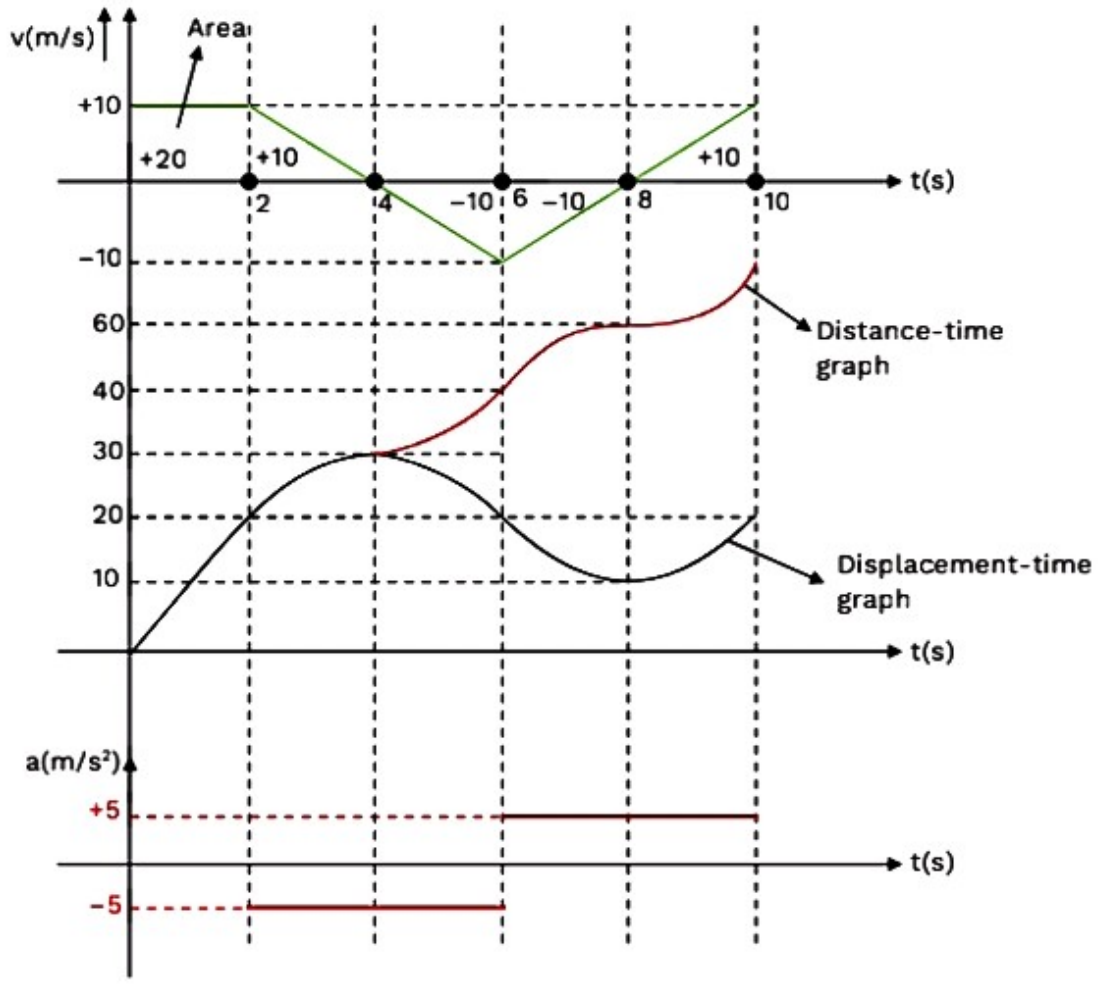
$$a = \frac{10 - 0}{2 - 0} = 5 \text{ m/s}^2$$

From t = 2 to t = 4 sec, a = 0
From t = 4 to t = 8 sec

$$a = \frac{0 - 10}{6 - 4} = -5 \text{ m/s}^2$$

Q. **For given velocity-time graph, draw a-t, displacement-t, distance-t graphs.**

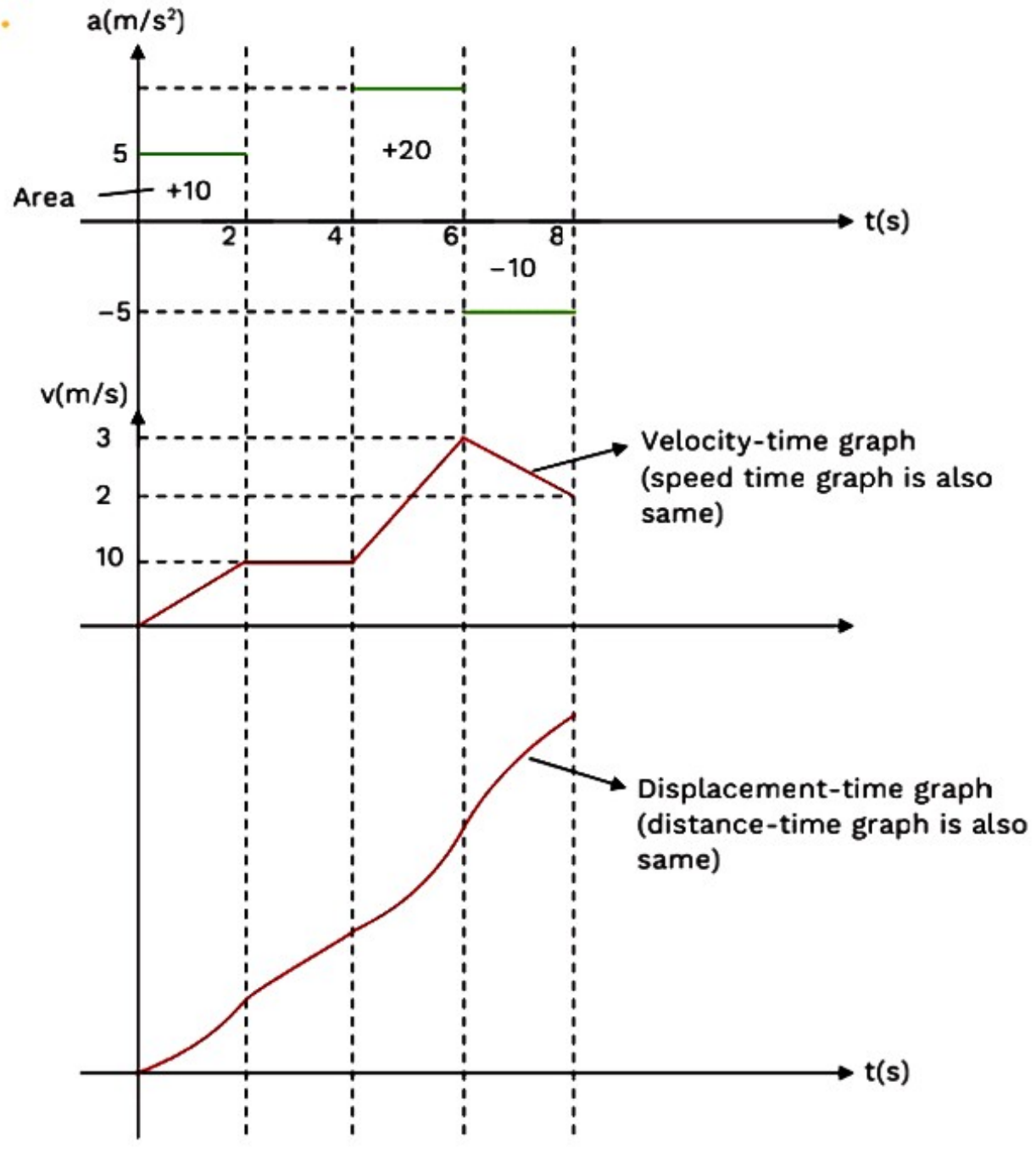
Sol.



Area under v-t curve gives displacement.

Q. **For given acceleration-time graph, draw v-t, displacement-time, distance-time graphs.**

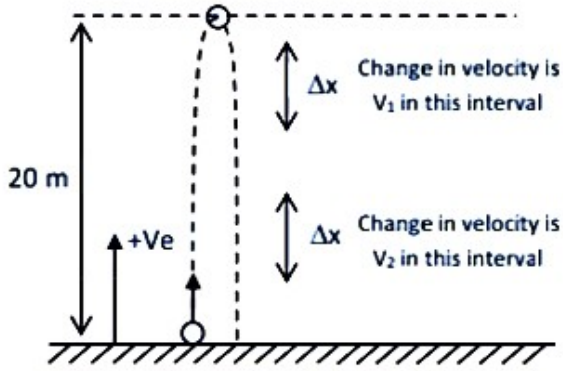
Sol.



Area under a-t graph gives change in velocity.

Here, Initial velocity = 0

Q. **A particle is thrown upward with speed 20 m/s. Taking vertically upward direction as positive, draw v-x (velocity displacement) graph, where displacement is measured from ground.**



Particle takes more time to travel in this interval because it is moving slow.

Particle takes less time to travel in this interval because it is moving fast.

Sol. $\Delta v = g\Delta t$

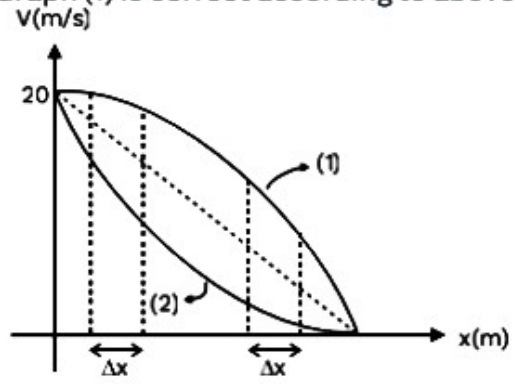
Δv will be more when Δt is more.

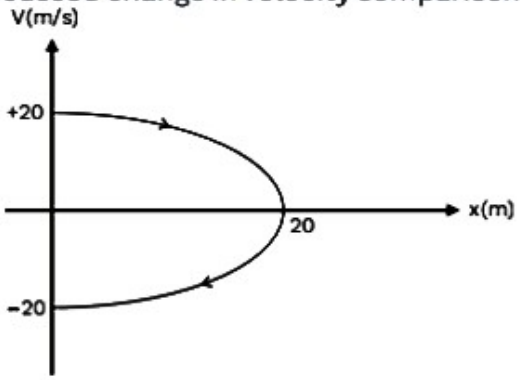
$\therefore \quad \Delta v_1 > \Delta v_2$

$\boxed{v^2 - u^2 = 2(-g)x}$ → Curve (parabola)

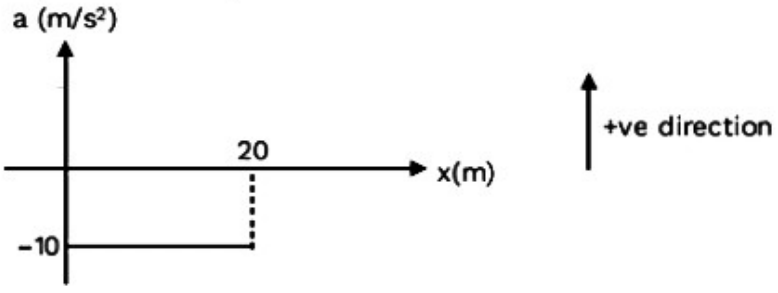
For upward motion

Graph (1) is correct according to above discussed change in velocity comparison.

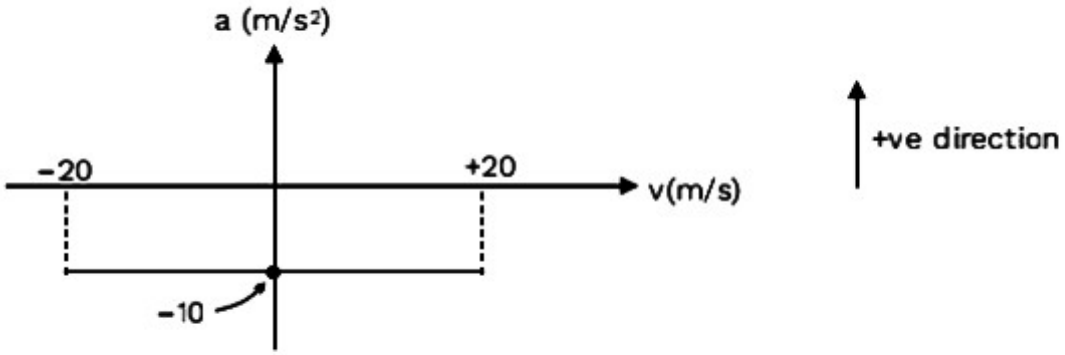




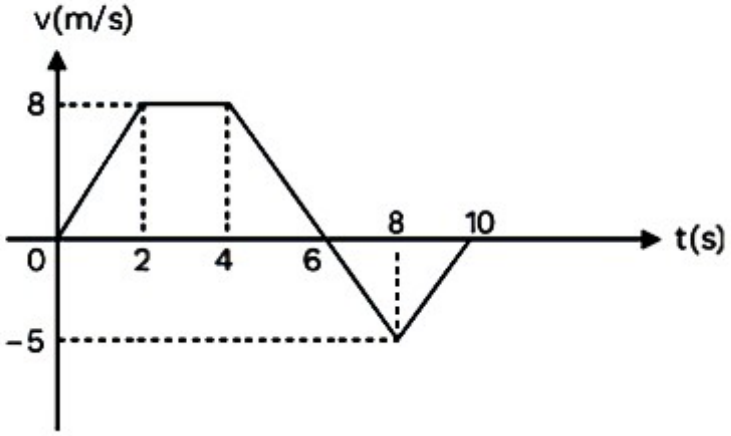
a-x graph for previous example



a-v graph for previous example



Q. **For a particle moving along x-axis, velocity-time graph is as shown in figure. Find the distance travelled and displacement of the particle ?**



Sol. Distance travelled = Area under v-t graph (taking all areas as +ve)

$$= \frac{1}{2}(2+6)\times 8 + \frac{1}{2}\times 4\times 5 = 32 + 10 = 42 \text{ m}$$

Displacement = Area under v-t graph (taking areas below time axis as -ve)

Displacement = Area of trapezium − Area of triangle

$$= \frac{1}{2}(2+6)\times 8 - \frac{1}{2}\times 4\times 5 = 32 - 10 = 22 \text{ m}$$

Motion in a Plane

Projectile Motion

Example of motion in two dimensions.

A ball is thrown at an angle 'θ' with horizontal

For $\theta = 0°, 90° \Rightarrow$ Straight line motion.

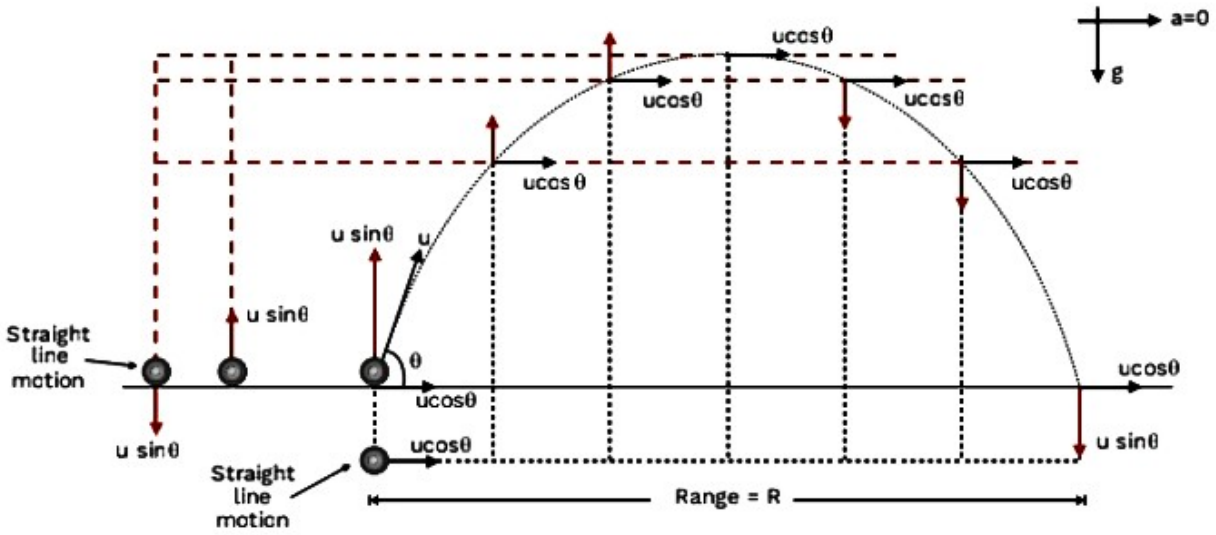
For $\theta \neq 0°$ or $90°$,

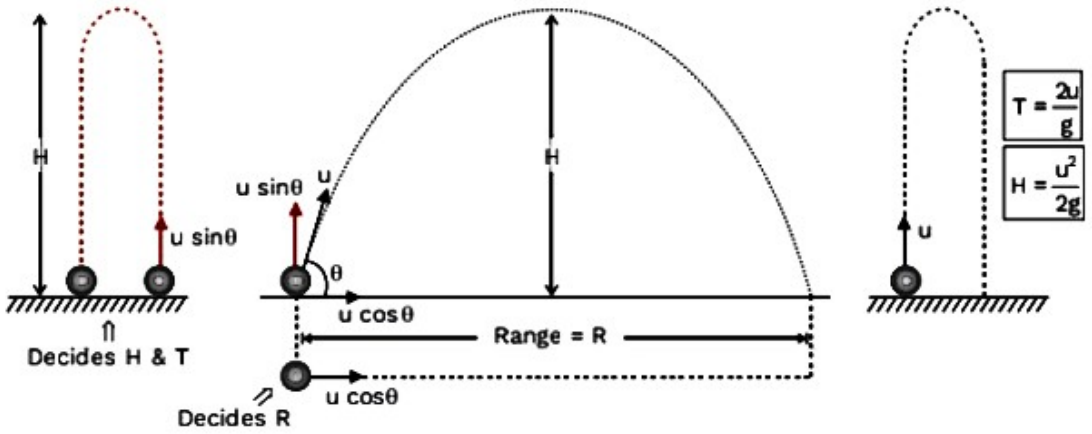
Maximum height = H

Time of flight = T

We can break the motion of ball is two straight lines

$\Rightarrow$ Two ball theory





Time of flight of projectile, $\boxed{T = \frac{2u\sin\theta}{g}}$

Maximum height (measured from ground), $\boxed{H = \frac{u^2 \sin^2\theta}{2g}}$

Horizontal Range, $R = u\cos\theta \times T$

$$R = u\cos\theta \times \frac{2u\sin\theta}{g}$$

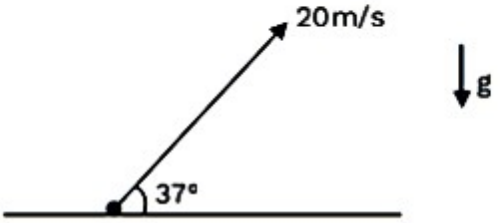
$$\boxed{R = \frac{2u^2 \sin\theta\cos\theta}{g} = \frac{u^2 \sin 2\theta}{g}}$$

Range is maximum when $\theta = 45°$

$$R_{max} = \frac{u^2 \sin(2\times 45°)}{g}$$

$$\boxed{R_{max} = \frac{u^2}{g}}$$

Q. **Find time of flight (T), maximum height (H) and range (R).**



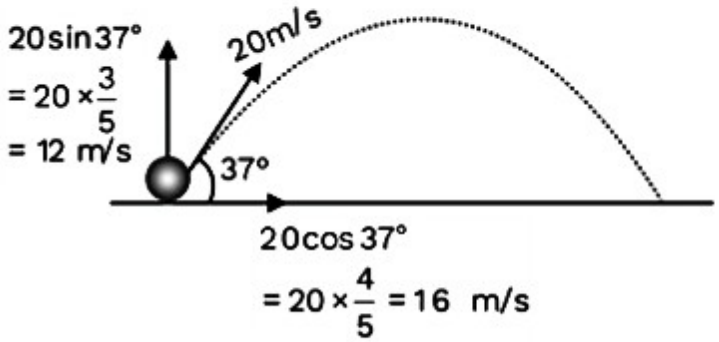
Sol.

$$T = \frac{2\times 12}{10} = 2.4 \text{ sec}$$

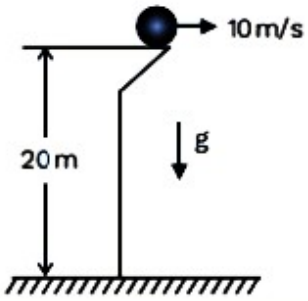
$$H = \frac{12\times 12}{2\times 10} = 7.2 \text{ m}$$

$$R = 16 \times 2.4 = 38.4 \text{ m}$$

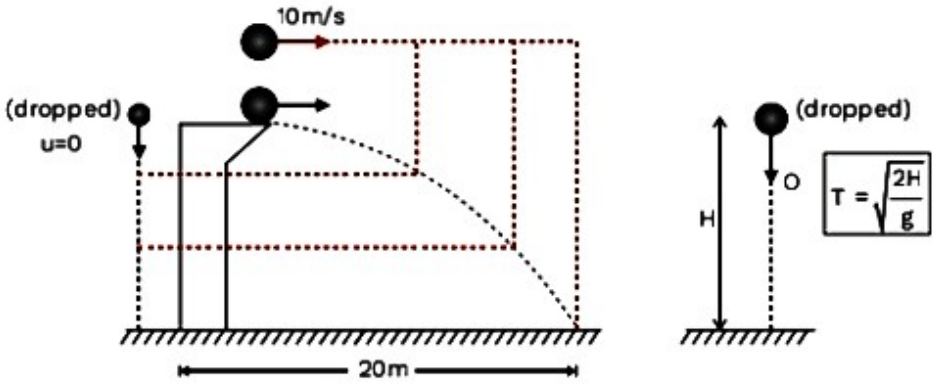
$$R = u\cos\theta \times T$$



Q. **Find time of flight and horizontal range.**



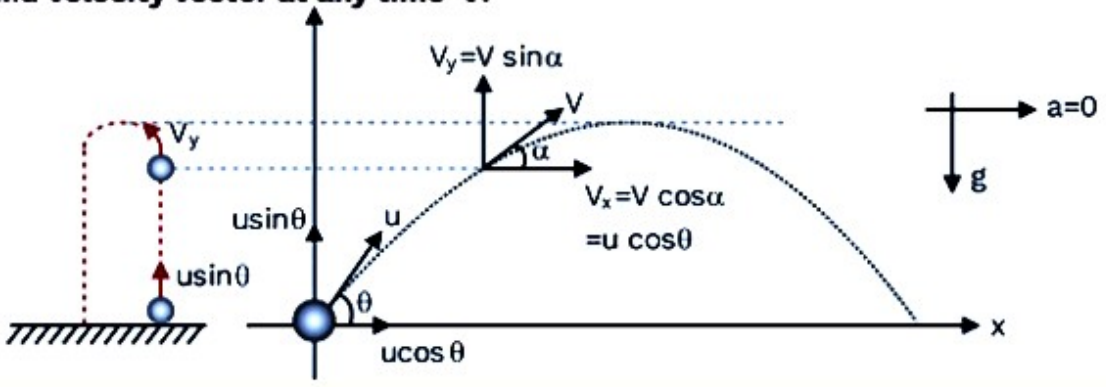
Sol. Apply two ball theory.



$$T = \sqrt{\frac{2 \times 20}{10}} = 2\,\text{sec}$$

Range, R = 10 × 2 = 20m

Q. **Find velocity vector at any time 't'.**



Sol. $V_x = V\cos\alpha = u\cos\theta$

$V = u + at$

$V_y = u\sin\theta + (-g)t = V\sin\alpha$

$\vec{V} = V_x\hat{i} + V_y\hat{j}$

$= V\cos\alpha\hat{i} + V\sin\alpha\hat{j}$

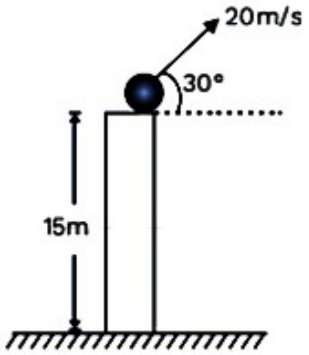
$\vec{V} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$

Q. **Find :**

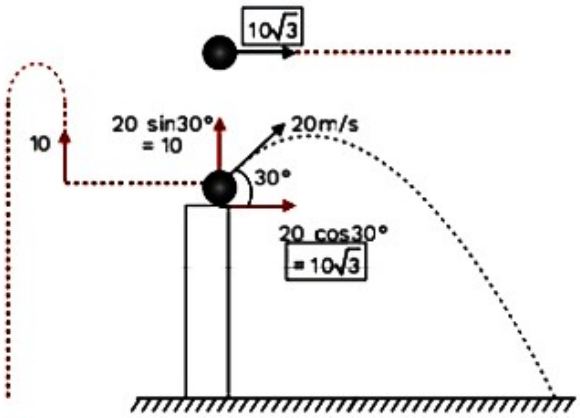
(1) Time of flight, T

(2) Maximum height attained by ball from ground, H

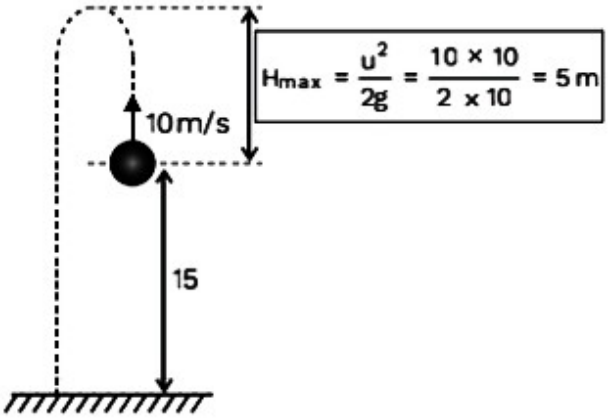
(3) Horizontal range, R



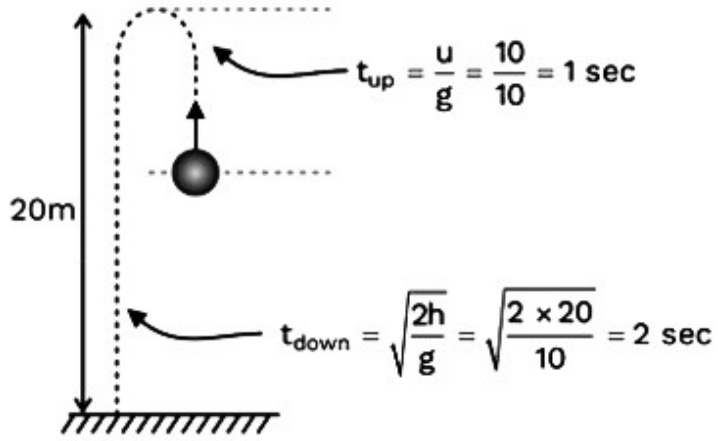
Sol.



For T and H



Maximum height attained by ball from ground, H = 15 + 5 = 20m



$\therefore$ Time of flight, $T = t_{up} + t_{down}$

$= 1 + 2 = 3$ sec.

Horizontal range, $R = 10\sqrt{3} \times 3 = 30\sqrt{3}\,m$

Time of flight can also be calculated by using

$S = ut + \frac{1}{2}at^2$ [consider vertical motion only]

$$-15 = 10t + \frac{1}{2}(-10)t^2$$

$$\Rightarrow -15 = 10t - 5t^2$$

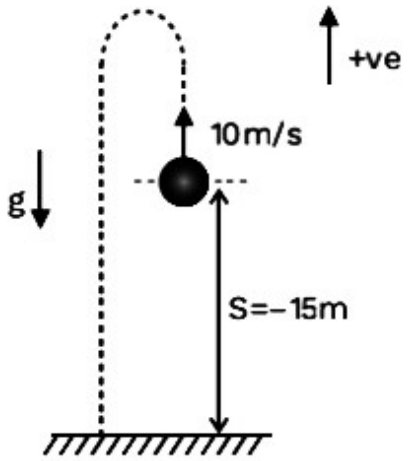
$$\Rightarrow -3 = 2t - t^2$$

$$\Rightarrow t^2 - 2t - 3 = 0$$

$$t = \frac{2 \pm \sqrt{4+12}}{2} = \frac{2 \pm 4}{2} = 1 \pm 2$$

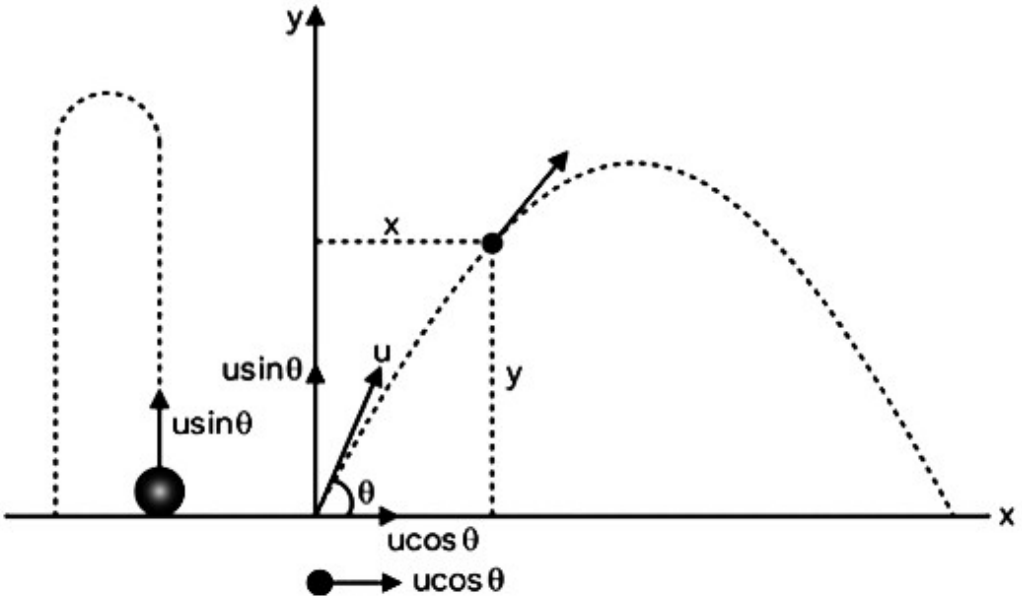
$$t = -1, \quad t = 3$$

$\therefore$ time of flight, T = 3 sec.



Equation of Trajectory (Equation of Path)

Relation between x and y co-ordinates.



A particle is projected at t = 0 from origin.

At t = t, particle is present at point (x, y)

$\Rightarrow x = u\cos\theta \times t$

$\Rightarrow y = u\sin\theta t - \frac{1}{2}gt^2$ → [Applying, $S = ut + \frac{1}{2}at^2$, $u = u\sin\theta$, $a = -g$]

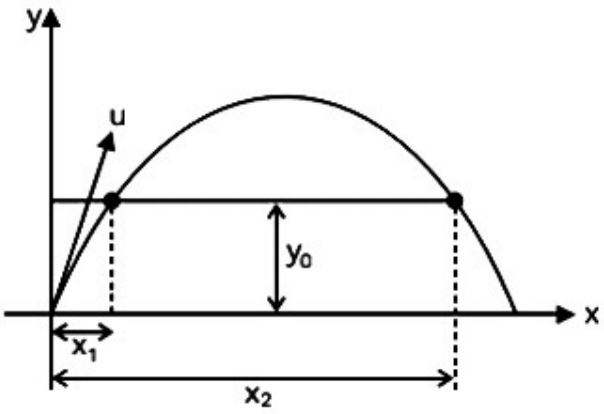
$\Rightarrow t = \frac{x}{u\cos\theta} \quad \Rightarrow y = u\sin\theta\frac{x}{u\cos\theta} - \frac{1}{2}g\frac{x^2}{u^2\cos^2\theta}$

$\Rightarrow y = \tan\theta x - \frac{1}{2}\frac{gx^2}{u^2\cos^2\theta}$ → [Path of particle is parabolic]

$y = ax - bx^2$ → Parabola

In cases of ground to ground projectile. For any value of y, we will get two values of x.

E.g. $\Rightarrow y = y_0 \begin{cases} \nearrow x = x_1 \\ \searrow x = x_2 \end{cases}$



Q. **Suppose a particle is thrown from origin with fixed speed 'u' at an angle 'θ' and passes through a fixed point (x_0, y_0). Find angle 'θ'.**

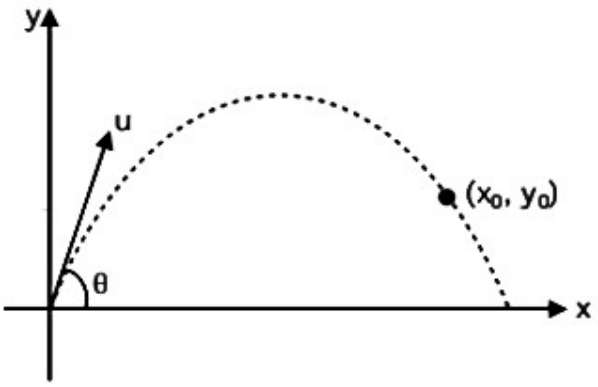
Sol.

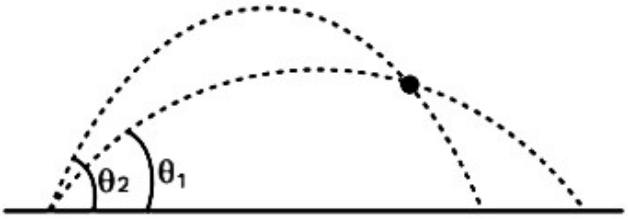
$$y = \tan\theta x - \frac{gx^2}{2u^2\cos^2\theta}$$

$$\Rightarrow y_0 = \tan\theta x_0 - \frac{gx_0^2(1+\tan^2\theta)}{2u^2}$$

$$\Rightarrow \frac{gx_0^2}{2u^2}\tan^2\theta - x_0\tan\theta + \left(y_0 + \frac{gx_0^2}{2u^2}\right) = 0$$

$$\Rightarrow \boxed{a\tan^2\theta - b\tan\theta + c = 0}$$

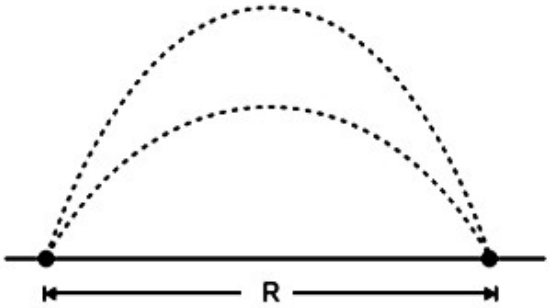




∴ We may get two angles of projection from above quadratic equation.

For example, we get same range for two angles of projection.

$\left.\begin{aligned}\theta_1 &= \theta \\ \theta_2 &= 90° - \theta\end{aligned}\right\}$ Same Range

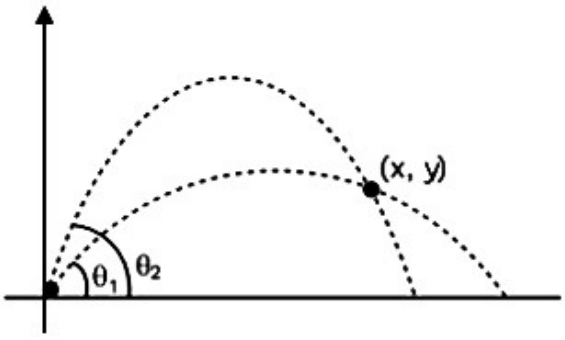


Range, $R = \frac{2u^2 \sin\theta\cos\theta}{g}$

$\theta_1 = \theta \Rightarrow R = \frac{2u^2 \sin\theta\cos\theta}{g}$

$\theta_2 = 90° - \theta \Rightarrow R = \frac{2u^2 \sin(90° - \theta)\cos(90° - \theta)}{g}$

$\Rightarrow R = \frac{2u^2 \cos\theta\sin\theta}{g} \Rightarrow$ Same Range



$y \neq 0$
$\theta_1 = \theta$
$\theta_2 \neq 90° - \theta$
$\theta_2 = \theta'$

Q. **A projectile projected from ground has velocity, $\vec{u} = \hat{i} + 2\hat{j}$. Find equation of trajectory.**

Sol.

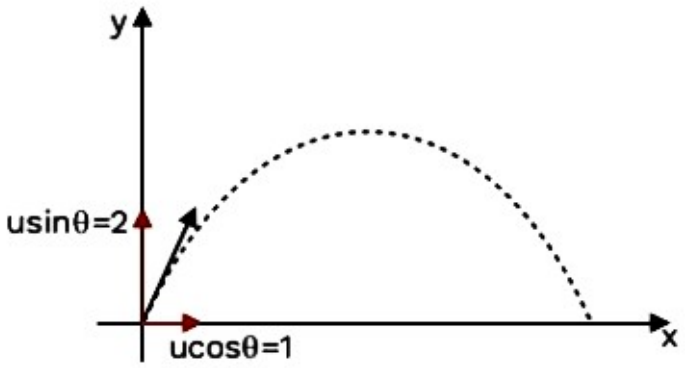
Equation of trajectory

$$y = \tan\theta \cdot x - \frac{gx^2}{2u^2\cos^2\theta}$$

$$= \frac{u\sin\theta}{u\cos\theta} \cdot x - \frac{g x^2}{2(u\cos\theta)^2}$$

$$y = \frac{2}{1}x - \frac{10}{2}\frac{x^2}{1^2}$$

$$= 2x - 5x^2$$



Q. **Equation of trajectory of a projectile is,** $y = 16x - \frac{5}{4}x^2$ **(x & y in m). Find Range.**

Sol.

$$y = \tan\theta \cdot x - \frac{gx^2}{2u^2\cos^2\theta}$$

$$y = 16x - \frac{5}{4}x^2$$

$$\boxed{\tan\theta = 16 = \frac{u\sin\theta}{u\cos\theta}} \qquad ...(1)$$

$$\frac{g}{2u^2\cos^2\theta} = \frac{5}{4} \Rightarrow u^2\cos^2\theta = 4 \Rightarrow \boxed{u\cos\theta = 2} \qquad ...(2)$$

From (1) and (2) $\Rightarrow u\sin\theta = 16 \times 2 = 32$

$$R = \frac{2u\sin\theta u\cos\theta}{g} = \frac{2 \times 32 \times 2}{10} = 12.8 \text{ m}$$

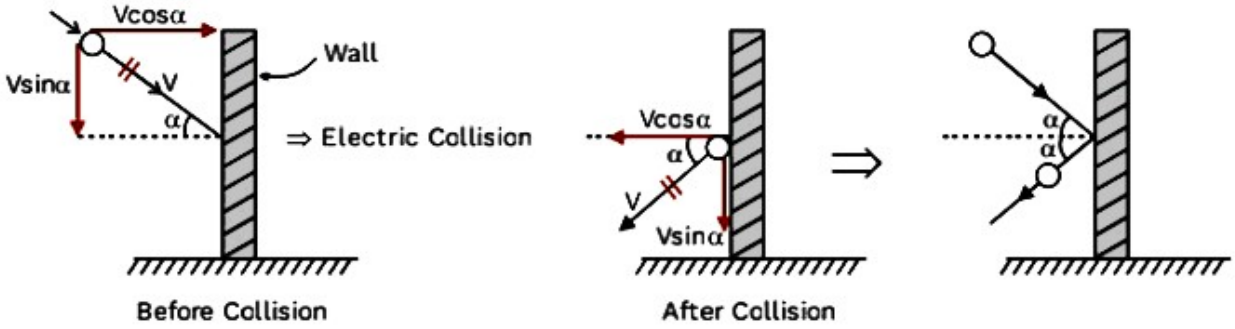
Other form of equation of trajectory.

$$y = \tan\theta \cdot x - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta\left(1 - \frac{gx^2}{2u^2\cos^2\theta x\tan\theta}\right)$$

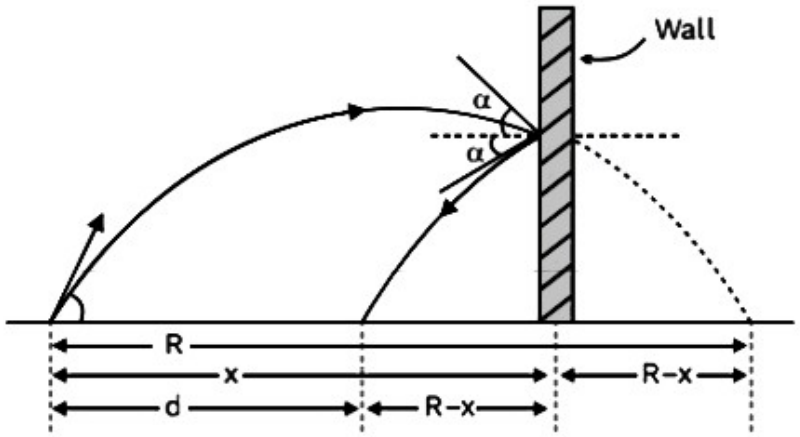
$$= x\tan\theta\left(1 - \frac{gx}{2u^2\cos^2\theta\frac{\sin\theta}{\cos\theta}}\right) = x\tan\theta\left(1 - \frac{gx}{2u^2\cos\theta\sin\theta}\right)$$

$$\boxed{y = x\tan\theta\left(1 - \frac{x}{R}\right)} \Rightarrow \text{Equation of trajectory (in other form)}$$

Collision of Projectile with a wall



Component of velocity which is normal to the wall has reversed its direction and component of velocity along the wall remains unchanged.



$d = x - (R - x)$

$d = 2x - R$

Average velocity in projectile motion :

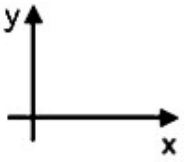
$$\vec{V}_{avg} = \frac{\text{Total displacement}}{\text{Total time}} = \frac{\vec{D}_{total}}{t_{total}}$$

Suppose projectile motion is taking place in x-y plane.

$\vec{D}_{total} = D_x\hat{i} + D_y\hat{j}$

$\vec{V}_{avg} = \left(\frac{D_x}{t}\right)\hat{i} + \left(\frac{D_y}{t}\right)\hat{j}$

$\vec{V}_{avg} = V_x\hat{i} + V_y\hat{j}$



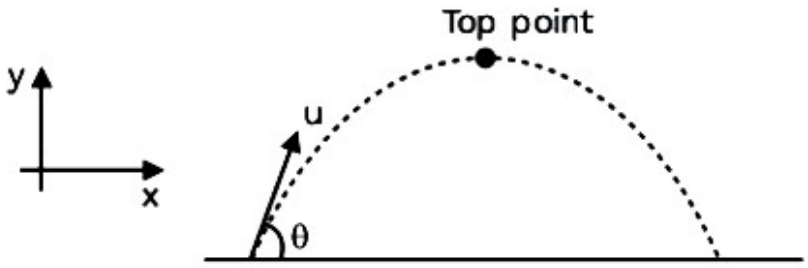
Q. **A particle is projected from ground with initial speed 'u' at an angle of projection 'θ' with horizontal. Find average velocity from,**

(a) Point of projection to top most point

(b) Top most point to landing point

(c) Point of projection to point of landing

Sol.



$$R = \frac{2u^2 \sin\theta\cos\theta}{g}$$

$$H = \frac{u^2 \sin^2\theta}{2g}$$

$$T = \frac{2u\sin\theta}{g}$$

(a) From point of projection to top most point

$$D_x = \frac{P}{2},\ D_y = H,\ t = \frac{T}{2}$$

$$\vec{V}_{avg} = \frac{Dx}{t}\hat{i} + \frac{Dy}{t}\hat{j} = \frac{\left(\frac{R}{2}\right)}{\left(\frac{T}{2}\right)}\hat{i} + \frac{(H)}{\left(\frac{T}{2}\right)}\hat{j}$$

$$\vec{V}_{avg} = \frac{R}{T}\hat{i} + \frac{2H}{T}\hat{j}$$

$$\text{Now, } R = u\cos\theta \times T \Rightarrow \frac{R}{T} = u\cos\theta$$

$$\frac{H}{T} = \frac{\frac{u^2\sin^2\theta}{2g}}{\frac{2u\sin\theta}{g}} = \frac{u\sin\theta}{4}$$

$$\vec{V}_{avg} = u\cos\theta\hat{i} + \frac{u\sin\theta}{2}\hat{j}$$

(b) From topmost point to landing point

$$D_x = \frac{R}{2},\ D_y = -H,\ t = \frac{T}{2}$$

$$\therefore \quad \vec{V}_{avg} = \frac{D_x}{t}\hat{i} + \frac{Dy}{t}\hat{j} = \frac{\left(\frac{R}{2}\right)}{\left(\frac{T}{2}\right)}\hat{i} + \frac{(-H)}{\left(\frac{T}{2}\right)}\hat{j}$$

$$\vec{V}_{avg} = \frac{R}{T}\hat{i} - \frac{2H}{T}\hat{j}$$

$$\vec{V}_{avg} = u\cos\theta\hat{i} - \frac{u\sin\theta}{2}\hat{j}$$

(c) From point of projection to point of landing

$$D_x = R, \; D_y = 0, \; t = T$$

$$\vec{V}_{avg} = \left(\frac{D_x}{T}\right)\hat{j} + \left(\frac{0}{T}\right)\hat{j}$$

$$\vec{V}_{avg} = u\cos\theta\hat{i}$$

As x-component of velocity of particle is always constant ($u\cos\theta$) during the motion, average velocity in x-direction is always same value ($u\cos\theta$) and it doesn't depend on time-interval taken.

Projectile from a moving platform

A particle is thrown at speed 'u' and angle 'θ' relative to a cart moving with speed 'v_0' on horizontal road as shown.

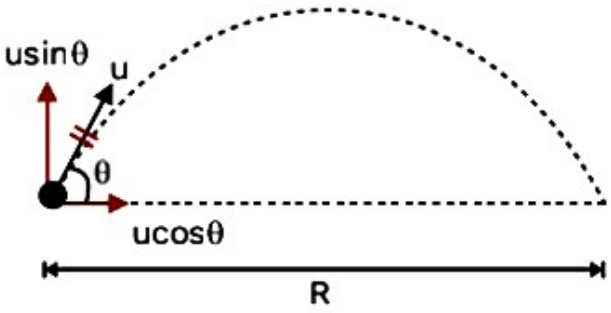
u = Initial speed of projectile relative to cart.

θ = Angle of projection relative to cart.

V_0 = speed of cart (relative to ground)

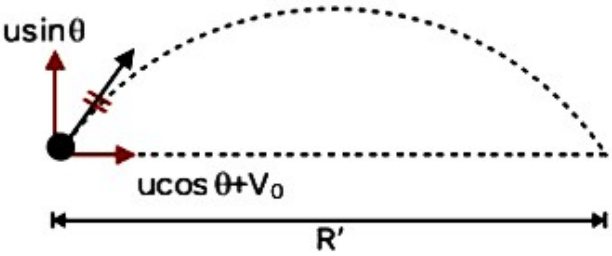


w.r.t cart



w.r.t ground

Relative to ground, ball has one more velocity due to motion of cart. Addition of both the velocities gives velocity of ball relative to ground.



R' = Range relative to ground

Q. **A projectile is thrown with 25 m/s at angle of 37° relative to a cart moving on horizontal road with speed 8 m/s as shown in figure.**

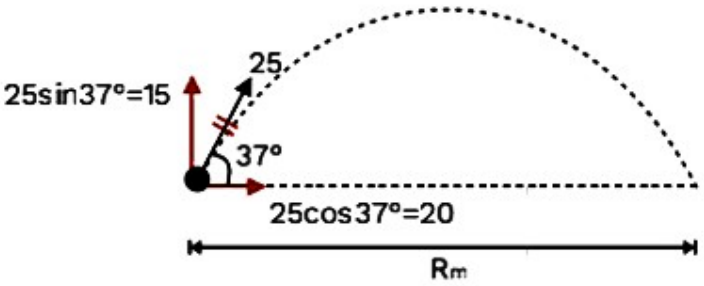


Find :

(a) Time of flight and range of projectile relative to man.

(b) time of flight and range of projectile relative to ground.

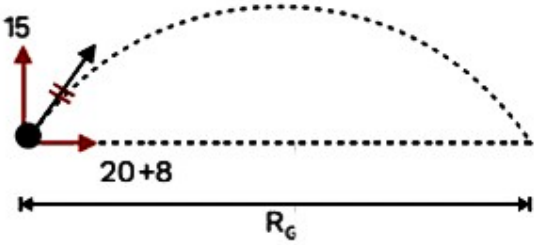
Sol. (a) Relative to man/cart



$T_m = \frac{2 \times 15}{10} = 3$ sec

$R_m = 20 \times 3 = 60$ m

(b) Relative to ground



$$T_G = \frac{2 \times 15}{10} = 3\,\text{sec}$$

$R_G = 28 \times 3 = 84$ m

Here, $T_m = T_G$, because there is no difference in vertical velocity and acceleration in both cases.

Also, $R_G - R_m = 84 - 60 = 24$ m.

Because in 3 seconds man (cart) will move by a distance of 24 m ($8 \times 3 = 24$ m) in horizontal direction.

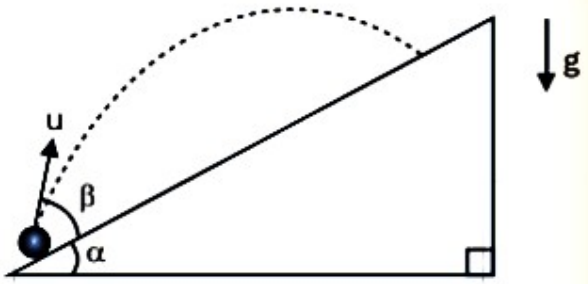
Projection on Incline Plane

Q. **An inclined plane makes an angle 'α' with horizontal. A projectile thrown with speed 'u' making an angle 'β' with incline surface as shown in figure.**

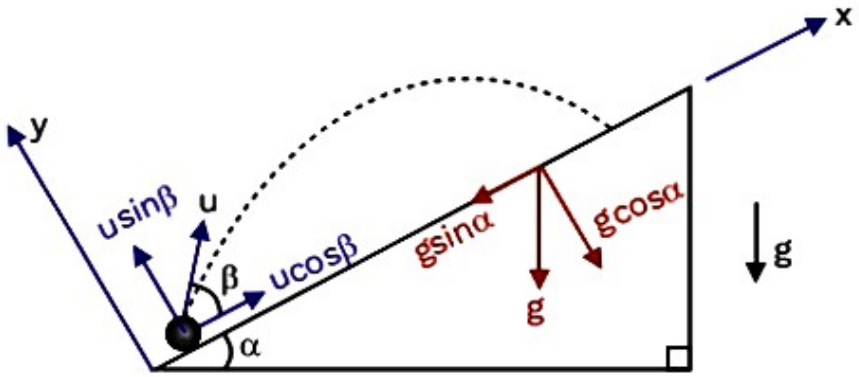
(i) Find maximum height attained by projectile measured from incline surface.

(ii) Find time of flight.

(iii) Find range along incline surface.



Sol.



By two ball concept.

Ball moving perpendicular to incline will give time of flight and maximum height.

Ball moving along the incline will give range along incline.

Time of flight, $\boxed{T = \frac{2u\sin\beta}{g\cos\alpha}}$

Maximum height, $\boxed{H = \frac{u^2\sin^2\beta}{2g\cos\alpha}}$

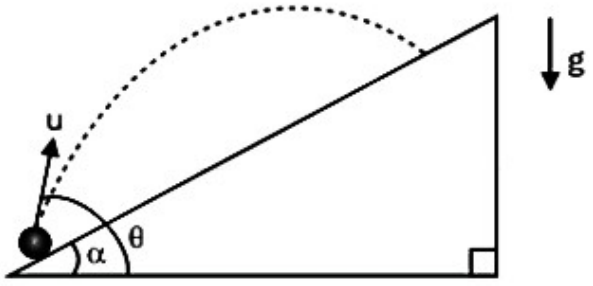
Along the incline: $S = ut + \frac{1}{2}at^2$

$\Rightarrow \boxed{R = u\cos\beta T - \frac{1}{2}g\sin\alpha T^2}$

Formula :

Up the Plane :

θ : Angle of projection with horizontal.



$\boxed{R = \frac{u^2}{g\cos^2\alpha}\left[\sin(2\theta - \alpha) - \sin\alpha\right]}$

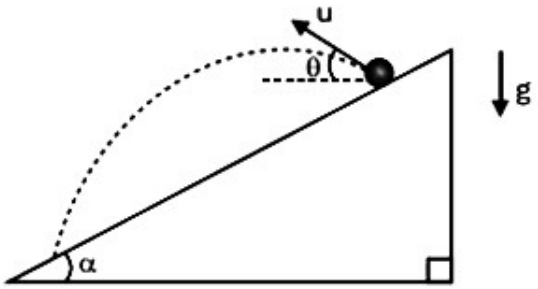
Range is maximum when $\boxed{\theta = \frac{\pi}{4} + \frac{\alpha}{2}}$

Down the Plane :

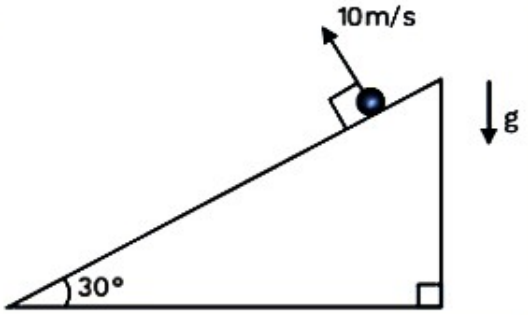
θ : Angle of projection with horizontal.

$$\boxed{R = \frac{u^2}{g\cos^2\alpha}\left[\sin(2\theta+\alpha)+\sin\alpha\right]}$$

Range is maximum when $\boxed{\theta = \frac{\pi}{4} - \frac{\alpha}{2}}$



Q. **A projectile is thrown perpendicularly to the incline.**
Find :
(1) Time of flight
(2) Maximum height
(3) Range



Sol. (1) $T = \frac{2\times 10}{5\sqrt{3}} = \frac{4}{\sqrt{3}}$ sec

(2) $H = \frac{10^2}{2\times 5\sqrt{3}} = \frac{10}{\sqrt{3}}$ m

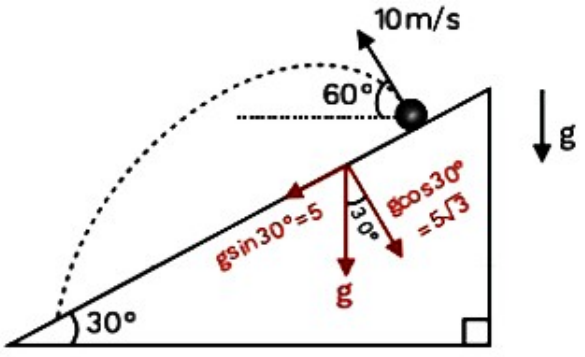
(3) $R = 0\times T + \frac{1}{2}\times 5\times T^2$

$= \frac{1}{2}\times 5\times\left(\frac{4}{\sqrt{3}}\right)^2 = \frac{40}{3}$ m

By formula : $\alpha = 30°$, $\theta = 60°$

$$R = \frac{100}{10\left(\frac{3}{4}\right)}\left[\sin(150°)+\sin(30°)\right]$$

$$= \frac{40}{3}\left[\frac{1}{2}+\frac{1}{2}\right] = \frac{40}{3}\text{ m}$$

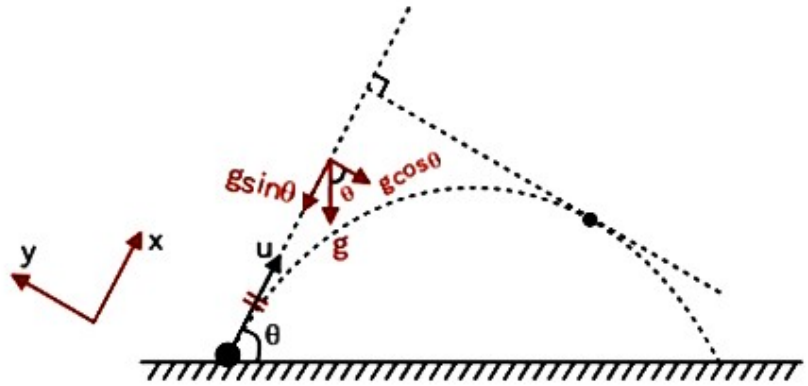


Q. **A particle is thrown with speed u making an angle 'θ' with horizontal.**

(1) Find the time after which velocity is perpendicular to initial velocity.

(2) Find the velocity at the instant it is perpendicular to initial velocity.

Sol.



In x-direction : $u_x = u,\ a_x = -g\sin\theta,\ v_x = 0$

In y-direction : $u_y = 0,\ a_y = -g\cos\theta,\ v_y = -v$

x-direction : $v_x = u_x + a_x t \Rightarrow 0 = u - g\sin\theta t$

$$\Rightarrow \boxed{t = \frac{u}{g\sin\theta}} \quad ...(1)$$

y-direction : $v_y = u_y + a_{yt}$

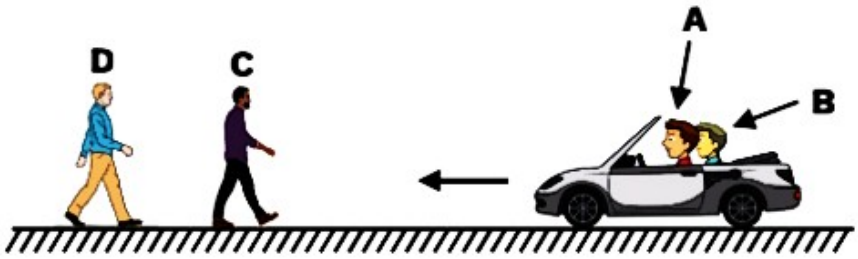
$$\Rightarrow -v = 0 - g\cos\theta \times \frac{u}{g\sin\theta}$$

$$\Rightarrow v = \frac{u\cos\theta}{\sin\theta} = u\cot\theta$$

$$\Rightarrow \boxed{v = u\cot\theta} \quad ...(2)$$

Relative Motion

Consider that two men A and B sitting in a moving car at constant speed are observed by two stationary men on ground.



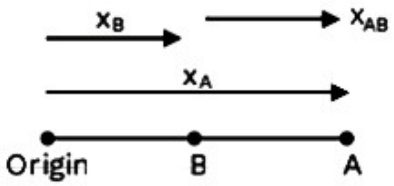
For the observers C and D, B appears to be in motion, but he appears to be at rest for A. Similarly C appears to be at rest for D but moving forward for A and B.

RELATIVE MOTION IN ONE DIMENSION

Relative Position :

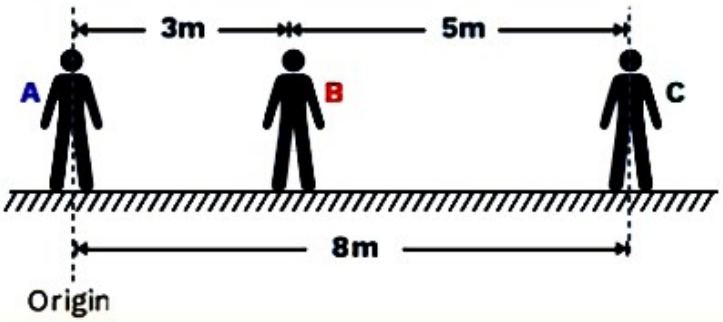
Position of the particle w.r.t observer is termed as relative postion.

In general, if the position of A w.r.t to origin is x_A and that of B w.r.t. origin is x_B then "Position of A w.r.t. B (x_{AB}) is given as,



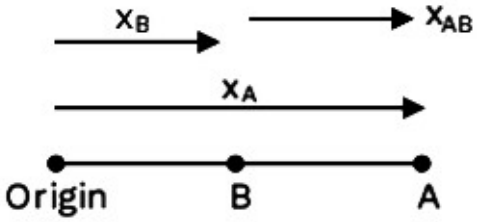
$$\boxed{x_{AB} = x_A - x_B}$$

Q. **In the given figure, take +ve direction towards right and -ve towards left. Find x_{BA}, x_{CA}, x_{CB}, x_{AB} and x_{AC}**



Sol. The position of B w.r.t A is 3m towards right $\Rightarrow$ $x_{BA} = +3m$
The position of C w.r.t A is 8m towards right $\Rightarrow$ $x_{CA} = +8m$
The position of C w.r.t B is 5m towards right $\Rightarrow$ $x_{CB} = +5m$
The position of A w.r.t B is 3m towards left $\Rightarrow$ $x_{AB} = -3m$
The position of A w.r.t C is 8m towards left $\Rightarrow$ $x_{AC} = -8m$
The position of B w.r.t C is 5m towards left $\Rightarrow$ $x_{BC} = -5m$

Relative Velocity in One Dimension :



If x_A is the position of A w.r.t. ground, x_B is position of B w.r.t. ground and x_{AB} is position of A w.r.t B then we can say

v_A = velocity of A w.r.t ground $= \frac{dx_A}{dt}$

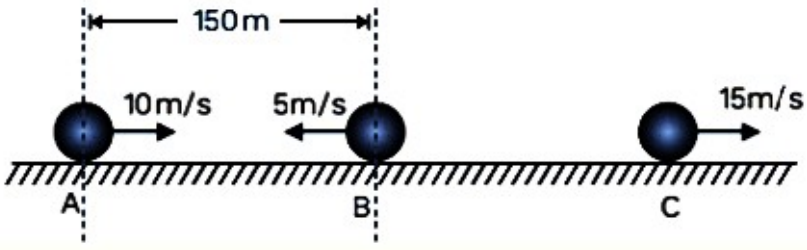
v_B = velocity of B w.r.t ground $= \frac{dx_B}{dt}$

and v_{AB} = velocity of A w.r.t B $= \frac{dx_{AB}}{dt} = \frac{d}{dt}(x_A - x_B) = \frac{dx_A}{dt} - \frac{dx_B}{dt}$

Thus

$$\boxed{v_{AB} = v_A - v_B}$$

Q. **In the given figure, three particles A, B and C are moving as shown. Separation between A and B is also given.**



Take positive direction towards right and negative towards left.
Find :

(a) V_{CA} **and** V_{AC}
(b) V_{BA} **and** V_{AB}
(c) V_{BC} **and** V_{CB}
(d) Velocity of approach between A and B
(e) Velocity of separation between A and C
(f) Time after which A and B will meet.

Sol. (a) Velocity of C w.r.t A,

$V_{CA} = V_C - V_A = 15 - 10 = 5$ m/sec

Velocity of A w.r.t C,

$V_{AC} = V_A - V_C = 10 - 15 = -5$ m/sec

$V_{CA} = -V_{AC}$

(b) Velocity of B w.r.t A,

$V_{BA} = V_B - V_A = (-5) - 10 = -15$ m/sec

$V_{AB} = V_A - V_B = 10 - (-5) = 15$ m/sec

(c) $V_{BC} = V_B - V_C = (-5) - (15) = -20$ m/sec

$V_{CB} = V_C - V_B = 15 - 10 = 5$ m/sec

(d) Velocity of approach is rate at which separation between two particles decreases.

Velocity of approach between A and B, $V_{app} = 10 + 5 = 15$ m/sec

(e) Velocity of separation is rate at which separation between two particles increase.

Velocity of separation between A and C, $V_{sep} = 15 - 10 = 5$ m/sec

(f) $$t = \frac{\text{Separation between A and B}}{V_{app} \text{ of A and B}} = \frac{150}{15} = 10 \text{ sec}.$$

Relative Acceleration :

The rate of change of relative velocity is called relative acceleration.

$$a_{AB} = \frac{dv_{AB}}{dt} = \frac{dv_A}{dt} - \frac{dv_B}{dt} = a_A - a_B$$

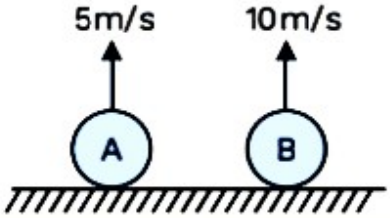
Equations of motion when relative acceleration is constant

$$V_{rel} = u_{rel} + a_{rel}\, t$$

$$s_{rel} = u_{rel} t + \frac{1}{2} a_{rel} t^2$$

$$v_{rel}^2 = u_{rel}^2 + 2 a_{rel}\, s_{rel}$$

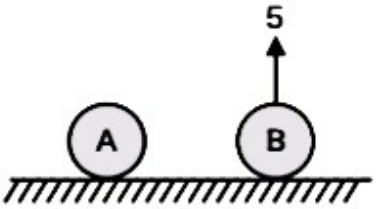
Q. **What is separation between two particles after 1 s when two particles A and B are shown vertically upwards with velocities 5 m/s and 10 m/s. Take g=10 m/s².**



Sol. $S_A = ut - \frac{1}{2}gt^2$

$= 5t - \frac{1}{2} \times 10 \times t^2 = 5 \times 1 - 5 \times 1^2 = 5 - 5 = 0$

$S_B = ut - \frac{1}{2}gt^2 = 10 \times 1 - \frac{1}{2} \times 10 \times 1^2 = 10 - 5 = 5$



$\therefore$ $S_B - S_A$ = separation = 5m.

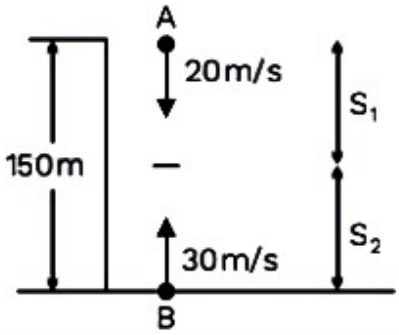
Aliter : $\vec{a}_{BA} = \vec{a}_B - \vec{a}_A = (-10) - (-10) = 0$

Also $\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = 10 - 5 = 5 \text{ m/s}$

$\therefore$ $\vec{S}_{BA}$ (in 1sec) $= \vec{v}_{BA} \times t = 5 \times 1 = 5$ m

$\therefore$ Separation between A and B after 1 sec = 5 m

Q. **From the top of a t.v. tower of height 150 m, a ball is thrown in vertically downward direction with speed 20 m/s and another ball is thrown in vertically upward direction with a speed 30 m/s from bottom of the building simultaneously. Find the time taken by the balls to meet. (g = 10 m/s²)**



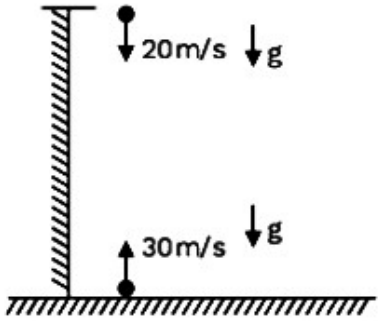
Sol. **Method-1**

$S_1 = 20\,t + 5\,t^2$

$S_2 = 30\,t - 5t^2$

$S_1 + S_2 = 150$

$\Rightarrow \quad 150 = 50\,t \Rightarrow t = 3\ s$



Method-2

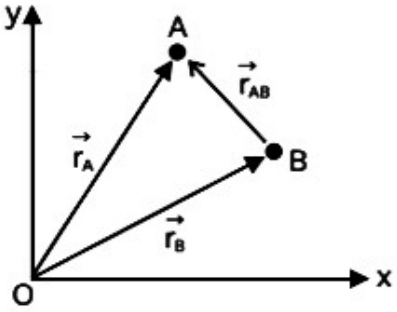
Relative acceleration of balls is zero as both balls have same acceleration in downward direction.

$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B = g - g = 0$

$\vec{v}_{BA} = 30 - (-20) = 50$

$s_{BA} = v_{BA} \times t$

$$t = \frac{S_{BA}}{V_{BA}} = \frac{150}{50} = 3\ s$$



Relative Motion in Two Dimension

$\vec{r}_A$ = positive of A with respect to O

$\vec{r}_B$ = position of B with respect to O

$\vec{r}_{AB}$ = position of A with respect to B

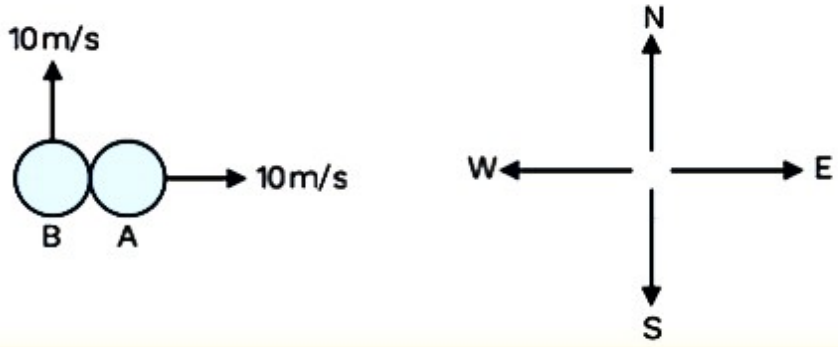
$\vec{r}_{AB} = \vec{r}_A - \vec{r}_B$ (The vector sum $\vec{r}_A - \vec{r}_B$ can be done

by Δ law of addition or resolution method)

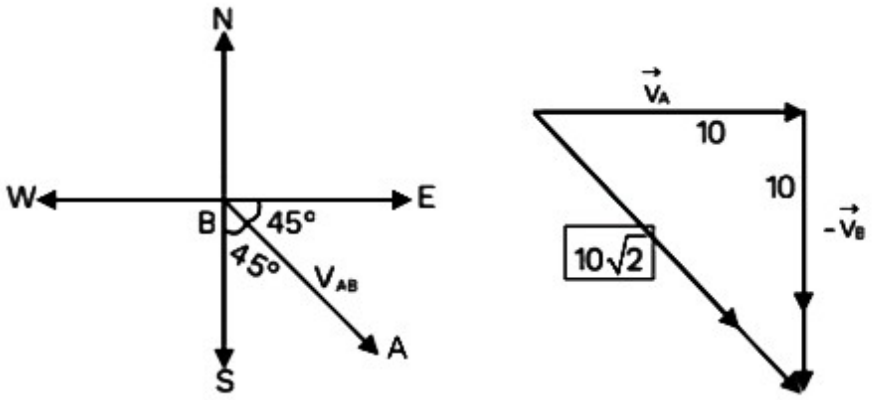
$$\therefore \quad \frac{d(\vec{r}_{AB})}{dt} = \frac{d(\vec{r}_A)}{dt} - \frac{d(\vec{r}_B)}{dt}$$

$$\Rightarrow \ \vec{v}_{AB} = \vec{v}_A - \vec{v}_B \ ; \ \frac{d(\vec{v}_{AB})}{dt} = \frac{d(\vec{v}_A)}{dt} - \frac{d(\vec{v}_B)}{dt} \Rightarrow \vec{a}_{AB} = \vec{a}_A - \vec{a}_B$$

Q. **What is relative velocity of A relative to B if object A and B are moving with same speed of 10 m/s, and the direction of motion of A is towards east while direction of motion of B is towards north starting from the same point as shown.**



Sol. **Method 1 :** $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ $\quad\therefore\quad |\vec{v}_{AB}| = 10\sqrt{2}$ m/s

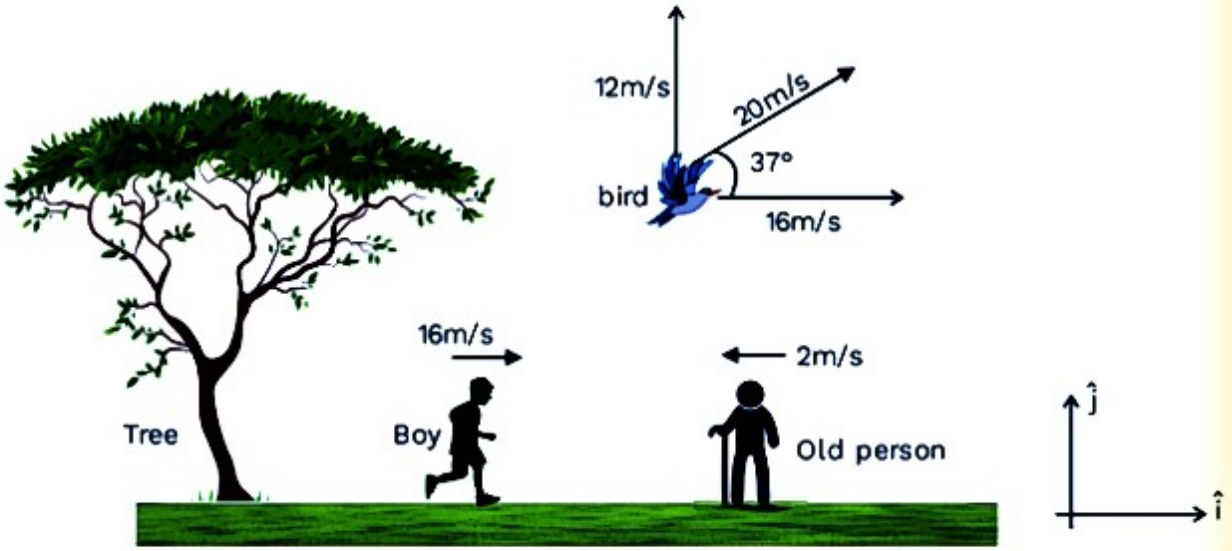


Method 2 : $\vec{v}_A = 10\hat{i}$, $\vec{v}_B = 10\hat{j}$

$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = 10\hat{i} - 10\hat{j}$

$\therefore |\vec{v}_{AB}| = 10\sqrt{2}$ m/s, direction of $\vec{v}_{AB}$ is south east.

Q. **A bird is flying over a boy and an old man moving towards each other as shown in the figure.**



(1) **What is the velocity of tree, bird and old man as seen by boy.**
(2) **What is the velocity of tree, bird and boy as seen by old man.**
(3) **What is the velocity of tree, boy and old man as seen by bird.**

Sol. (1) With respect to boy :

v_{Tree} = 16 m/s (←) or $-16\hat{i}$

v_{Bird} = 12 m/s (↑) or $12\hat{j}$

$v_{Old\ man}$ = 18 m/s (←) or $-18\hat{i}$

(2) With respect to old man :

V_{Boy} = 18 m/s (→) or $18\hat{i}$

v_{Tree} = 2 m/s (→) or $2\hat{i}$

v_{Bird} = 18 m/s (→) and 12 m/s (↑) or $18\hat{i} + 12\hat{j}$

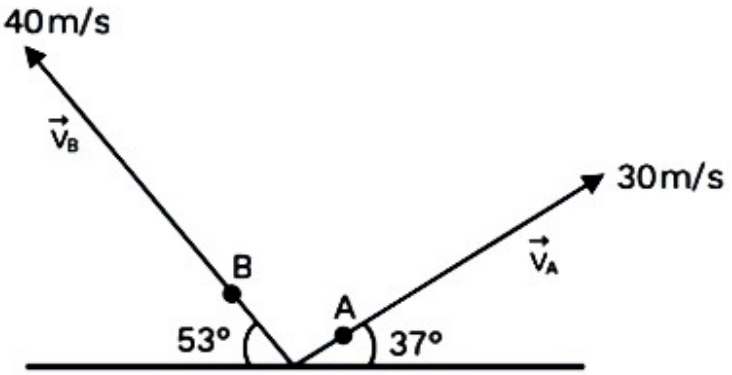
(3) With respect to Bird : v_{Tree} = 12 m/s (↓) and 16 m/s (←) or $-12\hat{j} - 16\hat{i}$

$v_{Old\ man}$ = 18 m/s (←) and 12 m/s (↓) or $-18\hat{i} - 12\hat{j}$

v_{Boy} = 12 m/s (↓) or $-12\hat{j}$

Relative Motion between Projectiles

Q. **Find the separation between the two particles after 1 s of throwing them in air as shown in the figure.**



Sol. $\vec{a}_{AB} = \vec{a}_A - \vec{a}_B = \vec{g} - \vec{g} = 0$

$\therefore \quad \vec{v}_{AB} = \sqrt{30^2 + 40^2} = 50$

$\therefore \quad s_{AB} = v_{AB}t = 50\,t = 50\,m$

RELATIVE MOTION IN A LIFT

Projectile Motion in a lift which is moving up with acceleration a

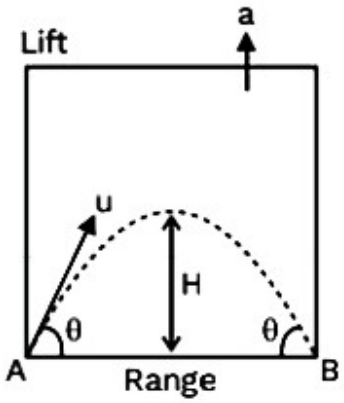
In the reference frame of lift, acceleration of a freely falling object is (g + a)

Velocity at maximum height = u cosθ

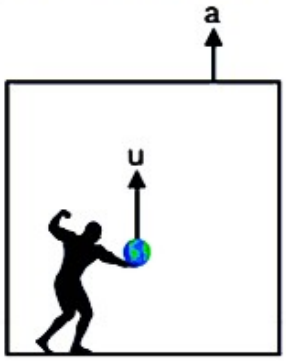
$$T = \frac{2u\sin\theta}{g+a}$$

Maximum height (H) $= \frac{u^2\sin^2\theta}{2(g+a)}$

Range $= \frac{u^2\sin 2\theta}{g+a}$



Q. **A lift is moving up with acceleration a. A person inside the lift throws the ball upwards with a velocity u relative to hand.**



(a) Calculate the flight time of ball.
(b) What is the maximum height reached by the ball in the lift ?

Sol. (a) $\vec{a}_{BL} = \vec{a}_B - \vec{a}_L = g + a$

$$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}_{BL}t^2$$

$$0 = uT - \frac{1}{2}(g + a)T^2 \quad \therefore \quad T = \frac{2u}{(g + a)}$$

(b) $v^2 - u^2 = 2as$

$$0 - u^2 = -2(g + a)H \Rightarrow H = \frac{u^2}{2(g + a)}$$

Relative Motion in River Flow

$\vec{v}_{mR}$ = velocity of man relative to river or velocity of man in still water

$\vec{v}_R$ = velocity of river relative to ground

$\vec{v}_m$ = velocity of man relative to ground

$\vec{v}_{mR} = \vec{v}_m - \vec{v}_R$ or $\vec{v}_m = \vec{v}_{mR} + \vec{v}_R$

If $\vec{v}_R = 0$, then $\vec{v}_m = \vec{v}_{mR}$ in words, velocity of man in still water = velocity of man w.r.t river

River Problem in One Dimension :

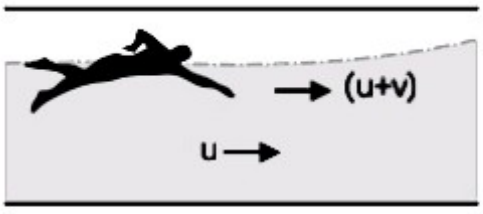
Velocity of river is u and velocity of man in still water is v.

Man swimming downstream (along the direction of river flow). In this case velocity of river

$v_R = + u$

Velocity of man w.r.t river $v_{mR} = +v$

Now $\vec{v}_m = \vec{v}_{mR} + \vec{v}_R = u + v$

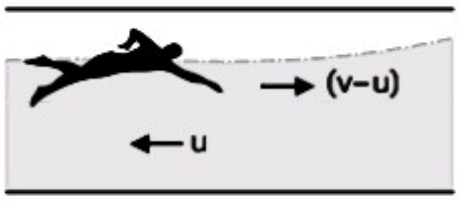


Man swimming upstream (opposite to the direction of river flow). In this case velocity of river

$\vec{v}_R = -u$

Velocity of man w.r.t river $\vec{v}_{mR} = +v$

Now $\vec{v}_m = \vec{v}_{mR} + \vec{v}_R = (v - u)$

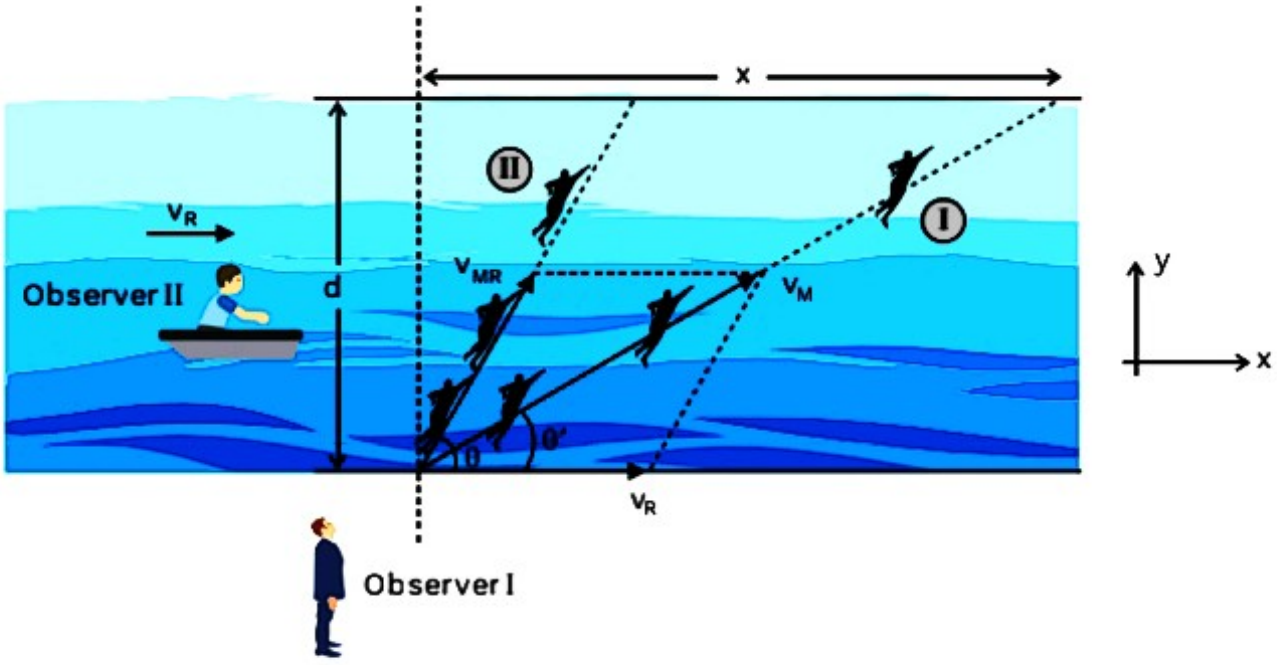


River problem in two dimension (crossing river) :

Consider a man swimming in a river with a velocity of $\vec{v}_{MR}$ relative to river at an angle of θ with the river flow. The velocity of river is $\vec{V}_R$. Let there be two observers I and II, observer I is on ground and observer II is on a raft floating along with the river and hence moving with the same velocity as that of river. Hence motion w.r.t observer II is same as motion w.r.t river. i.e., the man will appear to swim at an angle θ with the river flow for observer II.

For observer I the velocity of swimmer will be $\vec{v}_M = \vec{v}_{MR} + \vec{v}_R$,

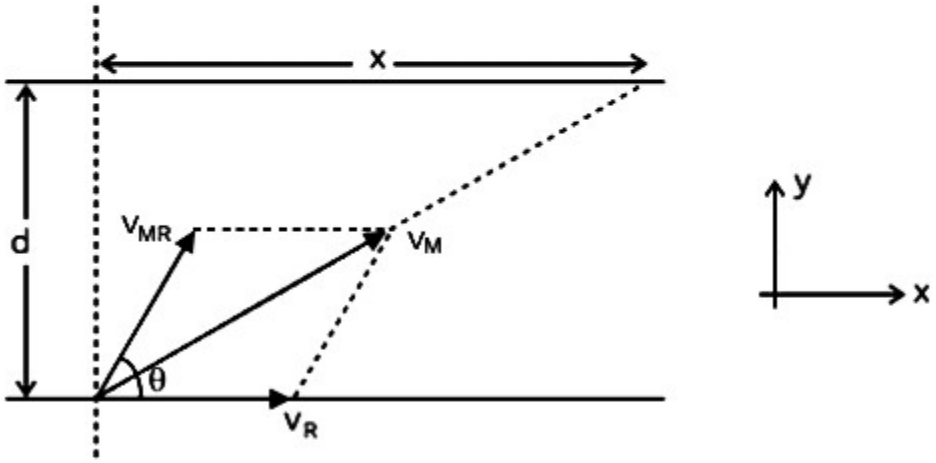
Hence the swimmer will appear to move at angle θ' with the river flow.



(1) : Motion of swimmer for observer I

(2) : Motion of swimmer for observer II

Consider a man swimming in a river with a velocity of $\vec{v}_{MR}$ relative to river at an angle of θ with the river flow. The velocity of river is V_R and the width of the river is d



$\vec{v}_M = \vec{v}_{MR} + \vec{v}_R \Rightarrow \vec{v}_M = (v_{MR}\cos\theta\hat{i} + v_{MR}\sin\theta\hat{j}) + v_R\hat{i} \Rightarrow \vec{v}_M = (v_{MR}\cos\theta + v_R)\hat{i} + V_{mR}\sin\theta\hat{j}$

Time to cross the river is t, then $t = \frac{d}{v_y} = \frac{d}{v_{MR}\sin\theta}$.

DRIFT

It is defined as the displacement of man in the direction of river flow. (See the figure).

If drift is x then,

Drift = $v_x \times t$

$$x = (v_{MR}\cos\theta + v_R) \times \frac{d}{v_{MR}\sin\theta}$$

Crossing the river in shortest time

$$t = \frac{d}{v_{MR}\sin\theta}$$

Clearly t will be minimum when $\theta = 90°$ i.e., time to cross the river will be minimum if man swims perpendicular to the river flow,

$$t_{min} = \frac{d}{v_{MR}}$$

Crossing the river in shortest path, Minimum Drift

When $v_{MR} > v_R$

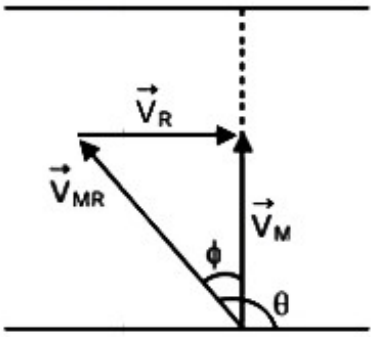
The minimum possible drift is zero. In this case the man swims in the direction perpendicular to the river flow as seen from the ground. This path is known as shortest path.

Here $x_{min} = 0 \Rightarrow (v_{MR}\cos\theta + v_R) = 0$ or $\cos\theta = -\frac{v_R}{v_{MR}}$

- Since $\cos\theta$ is -ve, $\therefore \theta > 90°$, i.e., for minimum drift the man must swim at some angle ϕ with the perpendicular in backward direction. Where $\sin\phi = \frac{v_R}{v_{MR}}$.

- $\theta = \cos^{-1}\left(\frac{-v_R}{v_{MR}}\right) \quad \therefore \left|\frac{v_R}{v_{MR}}\right| \le 1 \qquad$ i.e. $v_R \le v_{MR}$

i.e., minimum drift is zero if and only if velocity of man in still water is greater than or equal to the velocity of river.



- Time to cross the river along the shortest path

$$t = \frac{d}{v_{MR}\sin\theta} = \frac{d}{\sqrt{v_{MR}^2 - v_R^2}}$$

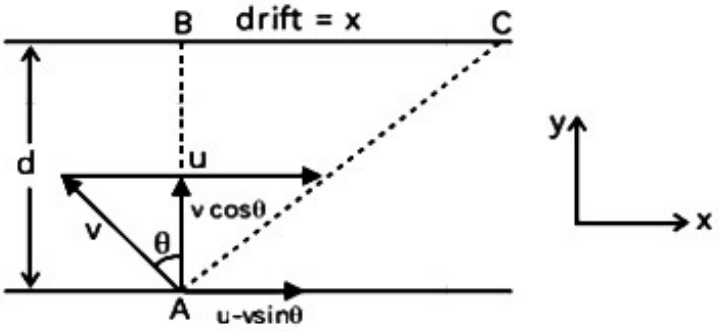
When $v_R > v_{MR}$

If $v_R > v_{MR}$ then it is not possible to have zero drift. In this case the minimum drift (corresponding to shortest possible path is non-zero.

Here, $v_R = u$ and $v_{MR} = v$

Time taken to cross the river is

$$t = \frac{d}{v_y} = \frac{d}{v\cos\theta}.$$



$$\text{Drift } x = (v_x)t = (u - v\sin\theta)\frac{d}{v\cos\theta}$$

$$= \frac{ud}{v}\sec\theta - d\tan\theta$$

The drift x is minimum, when $\frac{dx}{d\theta} = 0$,

or $\left(\frac{ud}{v}\right)(\sec\theta, \tan\theta) - d\sec^2\theta = 0$

or $\frac{u}{v}\sin\theta = 1$

or $\boxed{\sin\theta = \frac{v}{u}}$

So, for the minimum drift, the boat must move at an angle $\theta = \sin^{-1}\left(\frac{v}{u}\right)$ from normal direction.

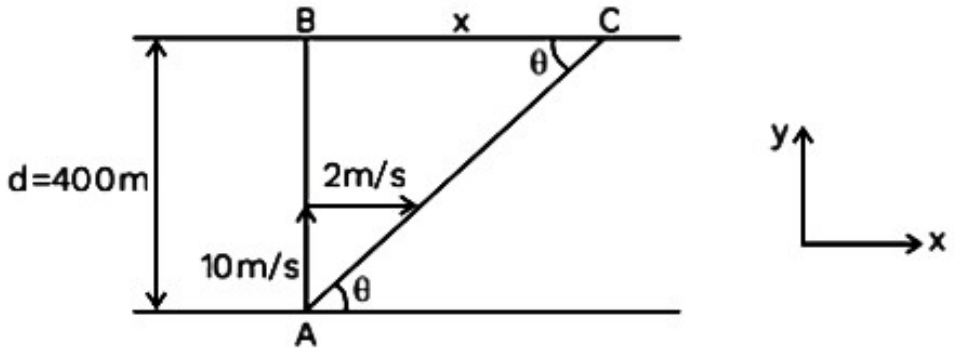
Q. **A 400m wide river is flowing at a rate of 2.0 m/s. A boat is sailing perpendicular to the river with a velocity of 10 m/s wrt water,**

(a) **What is the time the boat takes to reach the opposite bank?**

(b) **From the point directly opposite to the starting point, what is the distance of the final point of the boat after crossing the river?**

(b) **In what direction with the river flow, does the boat actually moves?**

Sol.



(a) time taken to cross the river $t = \frac{d}{v_y} = \frac{400\,m}{10\,m/s} = 40\,s$

(b) Drift (x) = $(v_x)(t)$ = (2 m/s) (40 s) = 80 m

(c) Actual direction of boat, $\theta = \tan^{-1}\left(\frac{10}{2}\right) = \tan^{-1} 5$, (downstream) with the river flow.

Q. **A man can swim at the rate of 5 km/h in still water. A 1km wide river flows at the rate of 3 km/h. The man wishes to swim across the river directly opposite to the starting point.**

(a) **Along what direction must the man swim ?**

(b) **What should be his resultant velocity ?**

(c) **How much time will he take to cross the river ?**

Sol. (a) The velocity of man with respect to river v_{mR} = 5 km/hr, this is greater than the river flow velocity, therefore, he can cross the river directly (along the shortest path). The angle of swim must be

$$\theta = \frac{\pi}{2} + \sin^{-1}\left(\frac{v_r}{v_{mR}}\right) = 90° + \sin^{-1}\left(\frac{v_r}{v_{mR}}\right) = 90° + \sin^{-1}\left(\frac{3}{5}\right) = 90° + 37°$$

= 127° w.r.t the river flow or 37° w.r.t perpendicular in backward direction

(b) Resultant velocity will be

$$v_m = \sqrt{v_{mR}^2 - v_R^2} = \sqrt{5^2 - 3^2} = 4 \text{ km/hr}$$

Along the direction perpendicular to the river flow.

(c) Time taken to cross the $t = \frac{d}{\sqrt{v_{mR}^2 - v_R^2}} = \frac{1 \text{ km}}{4 \text{ km/hr}} = \frac{1}{4}\text{h} = 15 \text{ min}$

Q. **A man wishing to cross a river flowing with velocity u jumps at an angle θ with the river flow.**

(i) **Find the net velocity of the man with respect to ground if he can swim with speed v in still water.**

(ii) **In what direction does the man actually move.**

(iii) **Find how far from the point directly opposite to the starting point does the man reach the opposite bank, if the width of the river is d. (i.e., drift)**

Sol. (i) $v_{MR} = v,\ v_R = u\ ;\ \vec{v}_M = \vec{v}_{MR} + \vec{v}_R$

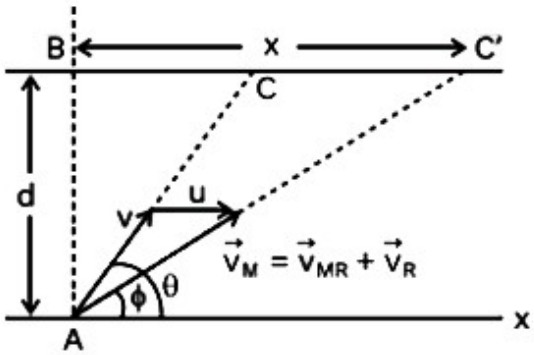
$\therefore$ Velocity of man,

$$v_M = \sqrt{u^2 + v^2 + 2vu\cos\theta}$$

(ii) $$\tan\phi = \frac{v\sin\theta}{u + v\cos\theta}$$

(iii) $$(v\sin\theta)t = d \Rightarrow t = \frac{d}{v\sin\theta}\ ;$$

$$x = (u + v\cos\theta)\,t = (u + v\cos\theta)\frac{d}{v\sin\theta}$$



Wind Airplane Problems :

Thus, velocity of aeroplane with respect to wind

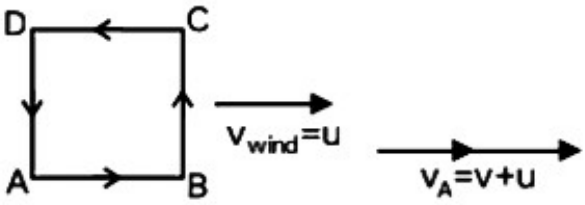
$$\vec{v}_{aw} = \vec{v}_a - \vec{v}_w \quad \text{or} \quad \vec{v}_a = \vec{v}_{aw} + \vec{v}_w$$

Where, $\vec{v}_a$ = velocity of aeroplane w.r.t ground and, $\vec{v}_w$ = velocity of wind.

Q. **Find the time an aeroplane having velocity v, takes to fly around a square with side a if the wind is blowing at a velocity u along one side of the square.**

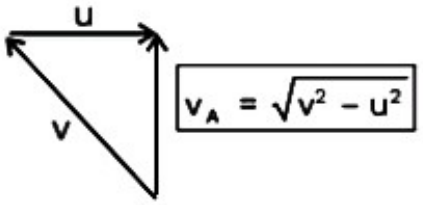
Sol. Velocity of aeroplane while flying through AB

$$v_A = v + u \; ; \; t_{AB} = \frac{a}{v+u}$$

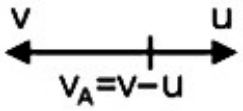


Velocity of aeroplane while flying through BC

$$v_A = \sqrt{v^2 - u^2} \; ; \; t_{BC} = \frac{a}{\sqrt{v^2-u^2}}$$

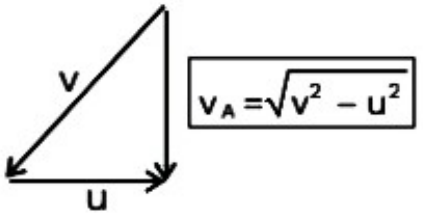


Velocity of aeroplane while flying through CD $v_A = v - u \; ; \; t_{CD} = \frac{a}{v-u}$



Velocity of aeroplane while flying through DA

$$v_A = \sqrt{v^2 - u^2} \; ; \; t_{DA} = \frac{a}{\sqrt{v^2-u^2}}$$



Total time $= t_{AB} + t_{BC} + t_{CD} + t_{DA}$

$$= \frac{a}{v+u} + \frac{a}{\sqrt{v^2-u^2}} + \frac{a}{v-u} + \frac{a}{\sqrt{v^2-u^2}} = \frac{2a}{v^2-u^2}\left(v + \sqrt{v^2-u^2}\right)$$

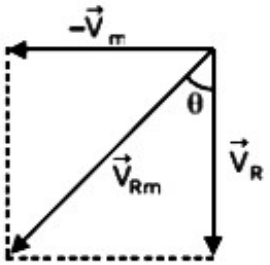
Rain Problem

If rain is falling vertically with a velocity $\vec{v}_R$ and an observer is moving horizontally with velocity $\vec{v}_m$, the velocity of rain relative to observer will be:

$$\vec{v}_{Rm} = \vec{v}_R - \vec{v}_m \quad \text{or} \quad v_{Rm} = \sqrt{v_R^2 + v_m^2}$$

and direction $\theta = \tan^{-1}\left(\frac{v_m}{v_R}\right)$ with the vertical as shown in figure.





Q. **Rain is falling vertically at speed of 10 m/s and a man is moving with velocity 6 m/s. Find the angle at which the man should hold his umbrella to avoid getting wet.**



Sol. $\vec{v}_{rain} = -10\hat{j} \Rightarrow \vec{v}_{man} = 6\hat{j}$

$\vec{v}_{r.w.r.t.\ man} = -10\hat{j} - 6\hat{i}$

$\tan\theta = \frac{6}{10} \Rightarrow \theta = \tan^{-1}\left(\frac{3}{5}\right)$

Where θ is angle with vertical

Q. **A man moving with 5 m/s observes rain falling vertically at the rate of 10 m/s. Find the speed and direction of the rain with respect to ground.**



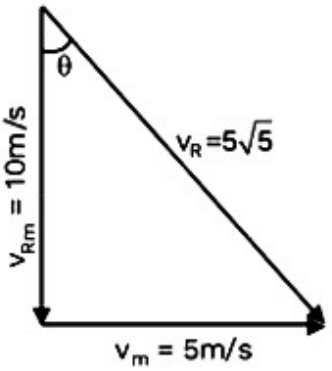
Sol. $v_{RM} = 10$ m/s, $v_M = 5$ m/s

$\vec{v}_{RM} = \vec{v}_R - \vec{v}_M$

$\Rightarrow \vec{v}_R = \vec{v}_{RM} + \vec{v}_M$

$\Rightarrow \vec{v}_R = 5\sqrt{5}$

$\tan\theta = \frac{1}{2}$, $\theta = \tan^{-1}\frac{1}{2}$



Q. **A standing man, observes rain falling with velocity of 20 m/s at an angle of 30° with the vertical.**

(i) **Find the velocity with which the man should move so that rain appears to fall vertically to him.**

(ii) **Now if he further increases his speed, rain again appears to fall at 30° with the vertical. Find his new velocity.**

Sol. (i) $\vec{v}_m = -v\hat{i}$ (Let)

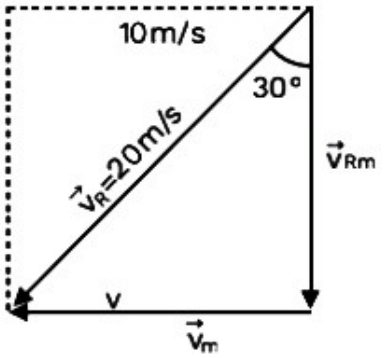
$\vec{v}_R = -10\hat{i} - 10\sqrt{3}\,\hat{j}$

$\vec{v}_{RM} = -(10\hat{i} - v) - 10\sqrt{3}\,\hat{j}$

$\Rightarrow \; -(10 - v) = 0$

(for vertical fall, horizontal component must be zero

or $v = 10$ m/s



(ii) $\vec{v}_R = -10\hat{i} - 10\sqrt{3}\,\hat{j}$

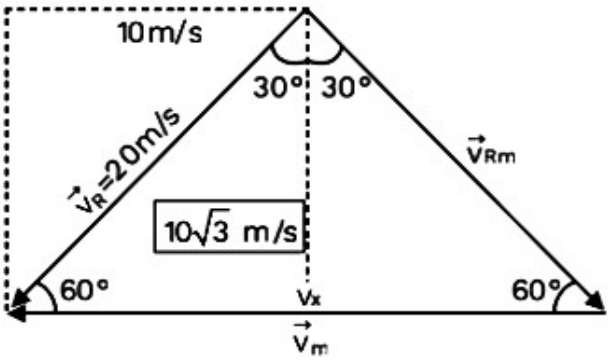
$\vec{v}_m = -v_x\hat{i}$

$\vec{v}_{RM} = -(10 - v_x)\hat{i} - 10\sqrt{3}\,\hat{j}$

Angle with the vertical = 30°

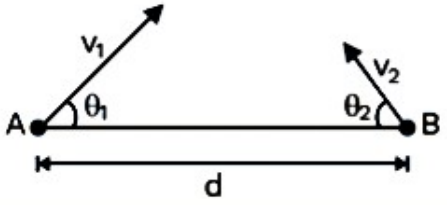
$\Rightarrow \; \tan 30° = \dfrac{10 - v_x}{-10\sqrt{3}}$

$\Rightarrow \; v_x = 20$ m/s



Velocity of Approach / Separation in Two Dimension

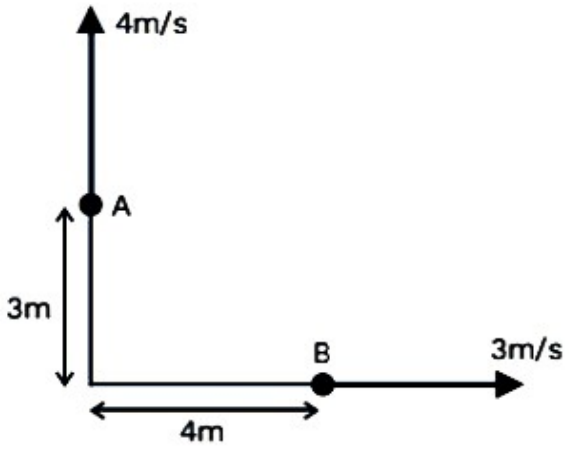
Q. **Two particles A and B are moving with constant velocities v_1 and v_2. At t = 0, v_1 makes an angle θ_1 with the line joining A and B and v_2 makes an angle θ_2 with the line joining A and B. find their velocity of approach.**



Sol. Velocity of approach is relative velocity along line AB

$v_{APP} = v_1 \cos\theta_1 + v_2 \cos\theta_2$

Q. **Particles A and B are moving as shown in the diagram at t = 0. Find their velocity of separation.**

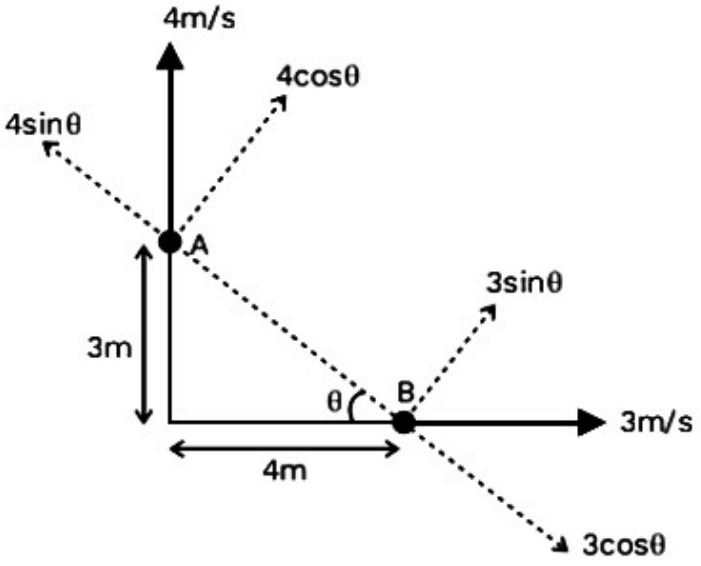


Sol. $\tan\theta = 3/4$

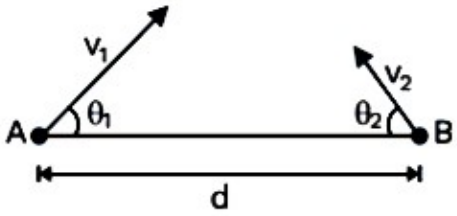
v_{sep} = relative velocity along line AB

$$= 3\cos\theta + 4\sin\theta$$

$$= 3\cdot\frac{4}{5} + 4\cdot\frac{3}{5} = \frac{24}{5} = 4.8 \text{ m/s}$$



Q. **Two particles A and B are moving with constant velocities v_1 and v_2. At t = 0, v_1 makes an angle θ_1 with the line joining A and B and v_2 makes an angle θ_2 with the line joining A and B.**



(i) **Find the condition for A and B to collide.**

(ii) **Find the time after which A and B will collide if separation between them is d at t = 0**

Sol. (i) For A and B to collide, their relative velocity must be directed along the line joining them.

Therefore their relative velocity along the perpendicular to this line must be zero.

Thus $v_1 \sin\theta_1 = v_2 \sin\theta_2$

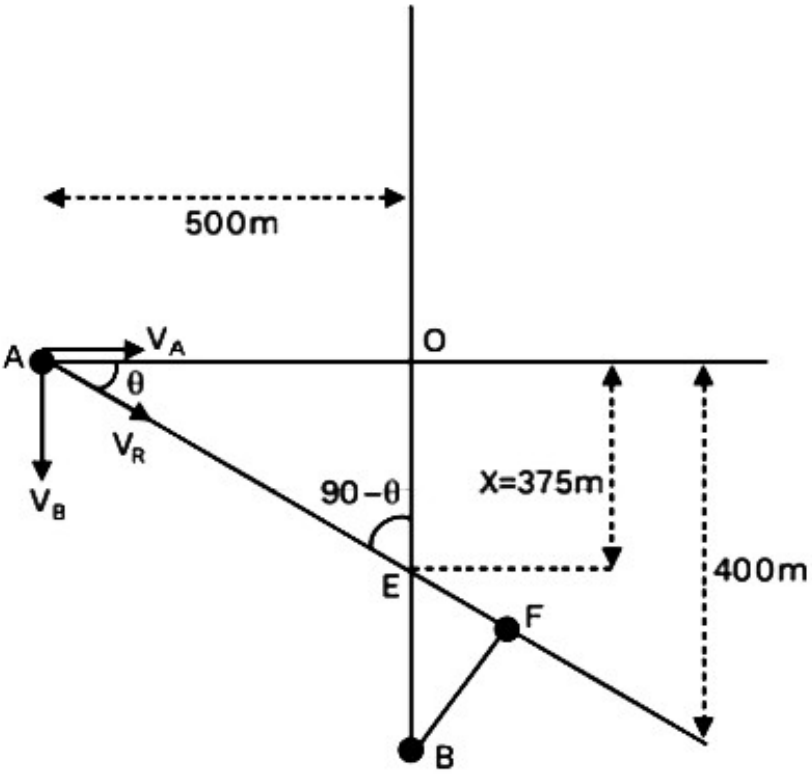
(ii) $v_{APP} = v_1 \cos\theta_1 + v_2 \cos\theta_2$; $t = \dfrac{d}{v_{app}} = \dfrac{d}{v_1 \cos\theta_1 + v_2 \cos\theta_2}$

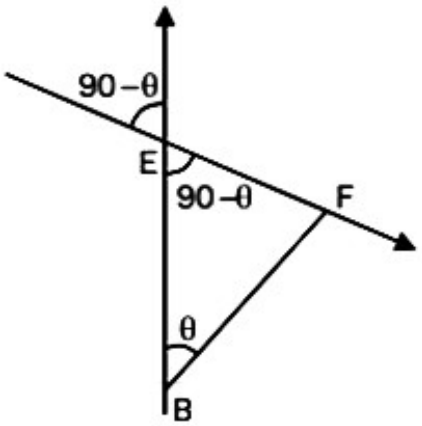
Minimum / Maximum distance between two particles

Q. **Two cars A and B are moving west to east and south to north respectively along crossroads. A moves with a speed of 72 kmh^{-1} and is 500 m away from point of intersection of cross roads and B moves with a speed of 54 kmh^{-1} and is 400 m away from point of intersection of cross roads. Find the shortest distance between them ?**

Sol. **Using the concept of relative velocity**

In this method we watch the velocity of A w.r.t B. To do this we plot the resultant velocity V_r. Since the accelerations of both the bodies is zero, so the relative acceleration between them is also zero. Hence the relative velocity will remain constant. So the path of A with respect to B will be straight line and along the direction of relative velocity of A with respect to B. The shortest distance between A and B is when A is at point F (i.e. when we drop a perpendicular from B on the line of motion of A with respect to B).





$$\tan\theta = \frac{V_B}{V_A} = \frac{15}{20} = \frac{3}{4} \qquad \text{...(i)}$$

This θ is the angle made by the resultant velocity vector with the x-axis.
Also we know that from figure

$$OE = \frac{x}{500} = \frac{3}{4} \qquad \text{....(ii)}$$

From equation (i) and (ii) we get, x = 375 m

∴ EB = OB – OE = 400 – 375 = 25 m

But the shortest distance is BF.

From magnified figure we see that BF = EB cos θ $= 25 \times \frac{4}{5}$

∴ BF = 20 m

Note :

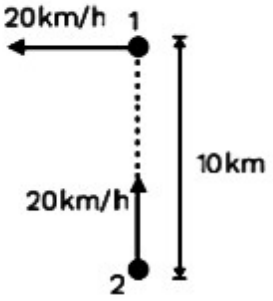
If the initial position of two particles are $\vec{r}_1$ and $\vec{r}_2$ and their velocities are $\vec{v}_1$ and $\vec{v}_2$ then shortest distance between the particles, $d_{shortest} = \frac{|\vec{r}_{12} \times \vec{v}_{12}|}{|\vec{v}_{12}|}$ and time after which this situation will occur, $t = -\frac{\vec{r}_{12} \cdot \vec{v}_{12}}{|\vec{v}_{12}|^2}$.

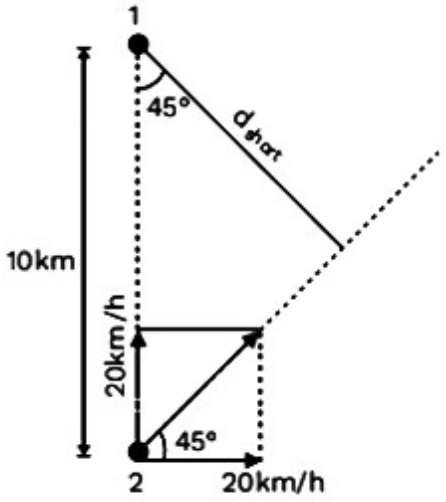
Q. **Two ships are 10 km apart on a line joining south to north. The one farther north is steaming west at 20 km h^{-1}. The other is steaming north at 20 km h^{-1}. What is their distance of closest approach? How long do they take to reach it?**

Sol. Solving from the frame of particle-1

We get $d_{short} = 10\cos 45° = \frac{10}{\sqrt{2}} = 5\sqrt{2}\,km$

$$t = \frac{10\sin 45}{|\vec{v}_{21}|} = \frac{10\times 1/\sqrt{2}}{20\sqrt{2}} = \frac{1}{4}h = 15\,min.$$





Q. **There are particles A, B and C are situated at the vertices of an equilateral triangle ABC of side a at t = 0. Each of the particles moves with constant speed v. A always has its velocity towards B, B towards C and C towards A. At what time will the particle meet each other ?**

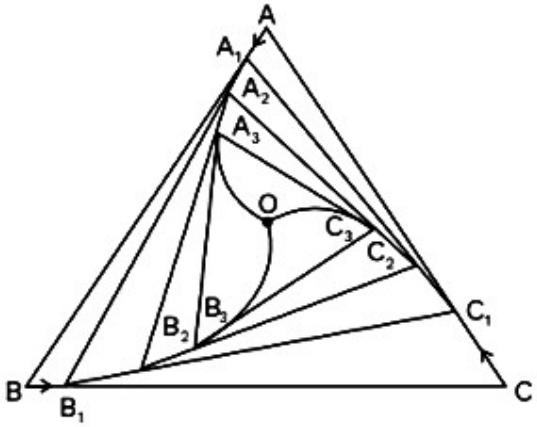
Sol. **Method-1**

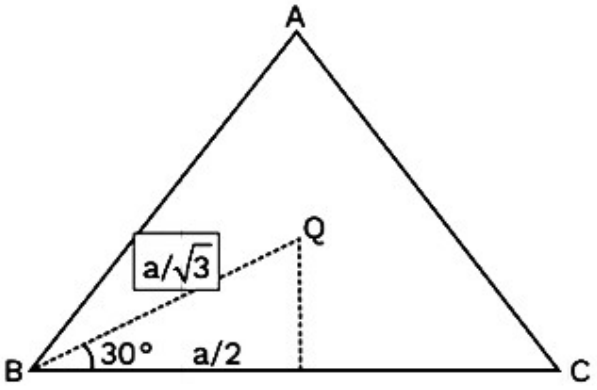
The motion of the particles is roughly sketched in figure. By symmetry they will meet at the centroid O of the triangle. At any instant the particles will from an equilateral triangle ABC with the same Centroid O. All the particles will meet at the centre. Concentrate on the motion of any one particle, say B. At any instant its velocity makes angle 30° with BO. The component of this velocity along BO is v cos 30°. This component is the rate of decrease of the distance BO. Initially

$$BO = \frac{a/2}{\cos 30^\circ} = \frac{a}{\sqrt{3}} = \text{displacement}$$

of each particle. Therefore, the time taken for BO to become zero.

$$= \frac{a/\sqrt{3}}{v\cos 30^\circ} = \frac{2a}{\sqrt{3}v \times \sqrt{3}} = \frac{2a}{3v}.$$



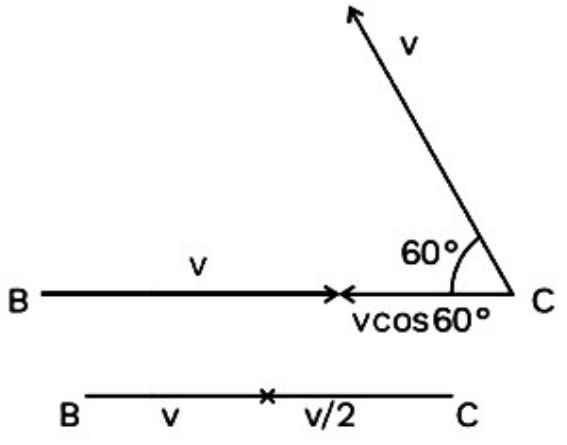


Method-2

Velocity of B is v along BC. The velocity of C is along CA. Its component along BC is v cos 60° = v/2. Thus, the separation BC decreases at the rate of approach velocity.

$\therefore$ approach velocity $= v + \frac{v}{2} = \frac{3v}{2}$

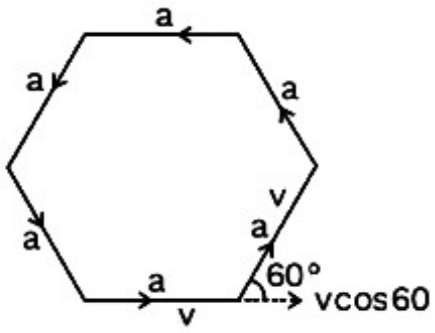
Since, the rate of approach is constant, the time taken in reducing the separation BC from a to zero is $t = \frac{a}{\frac{3v}{2}} = \frac{2a}{3v}$



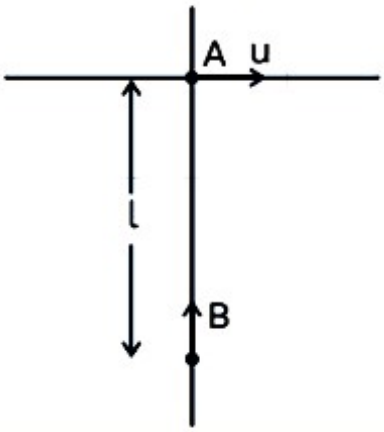
Q. **Six particles situated at the corners of a regular hexagon of side a move at a constant speed v. Each particle maintains a direction towards the particle at the next corner. Calculate the time the particles will take to meet each other.**

Sol. $V_{app} = V - V\cos 60° = V - V/2 = V/2$

$$t = \frac{a}{V_{app}} = \frac{a}{V/2} = \frac{2a}{V}$$



Q. **'A' moves with constant velocity u along the 'x' axis. B always has velocity towards A. After how much time will B meet A if B moves with constant speed v. What distance will be travelled by A and B.**



Sol. Let at any instant the velocity of B makes an angle α with that of x axis and the time to collide is T.

$$V_{app} = v - v\cos\alpha$$

$$\ell = \int_0^T V_{app}\,dt = \int_0^T (v - u\cos\alpha)\,dt \qquad ...(1)$$

Now equating the displacement of A and B along x direction we get

$$uT = \int v\cos\alpha\, dt \qquad \ldots(2)$$

Now from (1) and (2) we get

$$\ell = vT - \int_0^T u\cos\alpha\, dt = vT - \frac{u}{v}\int_0^T v\cos\alpha\, dt = vT - \frac{u}{v}\cdot uT$$

$$\Rightarrow T = \frac{\ell v}{v^2 - u^2}$$

Now distance travelled by A and B

$$= u \times \frac{\ell v}{v^2 - u^2} \text{ and } v \times \frac{\ell v}{v^2 - u^2} = \frac{uv\ell}{v^2 - u^2} \text{ and } \frac{v^2\ \ell}{v^2 - u^2}$$

